

BOILER FAILURE CASE STUDIES

Structured Root Cause Analysis of Boiler Emergencies, Protections and Pressure Part Failures

Atmospheric Fluidised Bed Combustion (AFBC) Captive Power Boiler

Eighteen classified failure cases, each analysed through asset context, incident timeline, evidence, hypothesis generation, hypothesis discrimination, failure mechanism, root cause and corrective action.

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1. Purpose and scope

This document sets out eighteen boiler failure cases in a common root cause analysis (RCA) format. Each case is analysed against the same eleven questions, so that the cases can be compared with one another and used as a reference set for incident investigation, operator training and maintenance strategy review.

The subject asset is a typical Indian captive-power Atmospheric Fluidised Bed Combustion (AFBC) boiler. The failure categories, causes, effects, protection settings and operator actions are drawn from the boiler emergency handling, boiler safety protection and boiler pressure part reference material supplied with this request.

2. How the cases were classified

The failures described in the source material fall naturally into six families. Each family is represented by one or more worked cases below:

- Water-side control failures - drum level low and drum level high, and the boiler feed pump failures that drive them.
- Steam-side control failures - main steam temperature high and low, main steam pressure high, and boiler outlet steam pressure low.
- Combustion and bed control failures specific to AFBC - bed temperature high (clinkering) and bed temperature low (bed slumping and unburnt accumulation).
- Furnace safety failures - furnace draught high, furnace explosion and flame failure.
- Pressure part failures - water wall / screen / evaporator tubes, superheater tubes, and air pre-heater (APH-PA and APH-SA) tubes.
- Auxiliary equipment failures - coal feeder (DCF) loss, PA line back-fire and hot bed material leakage, boiler feed pump failure, and loss of ID, FD, PA and SA fans.

3. Analysis method used in each case

Every case answers the same eleven questions in the same order. The logic runs from observation to mechanism to root cause:

- Evidence is listed in the order it became available, not in the order it was later understood, because the sequence of observation is itself diagnostic.
- Several competing hypotheses are generated before any is preferred, so that the analysis does not settle on the first plausible explanation.
- Each hypothesis is then tested against the evidence, and the single most discriminating item of evidence is identified - the observation that only one hypothesis can explain.
- Competing hypotheses are explicitly ruled out with stated reasoning, rather than being silently dropped.
- The failure mechanism describes the physical process by which the component failed. The root cause is obtained by asking why that mechanism was permitted to reach the point of failure - which almost always exposes a gap in the maintenance strategy, the operating envelope or the management system, not merely a broken part.

A note on the incidents: the cases are representative composites built from the causes, effects and actions given in the supplied reference material and from common AFBC operating experience. They are written as specific incidents because RCA reasoning cannot be demonstrated in the abstract, but they should be treated as worked examples for training and method, not as the verified record of any particular plant event.

4. Summary of cases

No.	Failure case (classification)	Function lost	Determined failure mechanism
1	Drum level low - water side	Feed water inventory control	Control valve actuator seizure
2	Drum level high - water side	Carry-over into SH and turbine	Instrument dynamic response failure
3	Steam temperature - high	Steam temperature control	Heat-input / attemperation imbalance
4	Steam temperature - low	Wet steam and water hammer	Spray valve seat leakage
5	Steam pressure - high	Pressure relief on load rejection	Relief path fail-to-open (hidden failure)
6	Steam pressure - low	Fuel delivery deficit	Coal hang-up and DCF shear pin failure
7	Bed temperature - high	Bed temperature control (AFBC)	Bed sintering / clinker formation
8	Bed temperature - low	Combustion window (AFBC)	Bed over-cooling by excess primary air
9	Furnace draught / explosion	Furnace safety	Deflagration after defeated purge
10	Flame failure	Light-up system reliability	Degraded atomisation and detector fouling
11	Pressure part - evaporator	Water wall / screen tube integrity	Fireside erosion to rupture
12	Pressure part - superheater	Superheater tube integrity	Creep rupture from internal deposits
13	Pressure part - APH	Air pre-heater integrity	Cold-end acid dew point corrosion
14	Fuel feed - electrical	Coal feeder availability	Common-cause MCC single-phasing
15	PA line back-fire	Primary air distribution	Branch air velocity loss and back-flow
16	Boiler feed pump	Feed water supply	Cavitation from NPSH loss
17	ID fan loss	Draught system	Bearing failure from loss of cooling water
18	PA fan loss	Fluidising air system	Motor cooling fouling plus hidden standby failure

Case 1 - Drum Level Low leading to water-wall starvation

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attemperator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. Two motor-driven BFPs (1W+1S) with auto-changeover on discharge header pressure; three-element drum level control with single-element fallback below 30% load.

2. Incident summary - what happened

During a night shift at 92% MCR, the drum level began falling steadily from 50% without any corresponding change in steam demand. The operator observed the feed control valve in DCS showing 100% open while feed flow remained lower than steam flow. Level touched the low alarm (20%), then the low-low pre-trip band; the running BFP tripped on low suction / recirculation and the standby did not pick up on auto. The boiler was tripped manually on drum level low-low before the water wall reached prolonged starvation. Post-trip inspection found a partially opened continuous blow-down (CBD) valve and a jammed feed control valve actuator.

2.1 Timeline and onset

- T-0:00 - Level 50%, steam flow 60 TPH, feed flow 60 TPH, stable.
- T+0:12 - Level drifting down at approx 1% per minute; feed flow reads 4-5 TPH less than steam flow (persistent negative mismatch).
- T+0:25 - Drum level Low alarm at 20%; FCV shows 100% demand, no further increase in feed flow.
- T+0:31 - BFP discharge header pressure dips; standby BFP fails to start on auto.
- T+0:34 - Level entering low-low band; superheater metal temperatures start rising, furnace draught steady (no tube leak signature).
- T+0:36 - Manual trip of boiler on drum level low-low; MS stop valve closed, feeding of all compartments cut.

3. Evidence, in order of observation

- Sustained negative feed-flow minus steam-flow mismatch with no load change - the earliest and most objective indication.
- Falling drum level with feed control valve demand saturated at 100%.
- BFP discharge header pressure lower than normal for the flow being delivered.
- No hissing / abnormal sound from the furnace and no rise in make-up water beyond the mismatch already explained by CBD.
- Boiler water conductivity rising rather than dropping (concentrating, not diluting).
- Flue gas temperature and furnace draught unchanged - no wet-steam signature.
- Standby BFP failed to start on auto command (auto-start relay picked up, breaker did not close).
- Physical inspection: CBD isolating valve found approximately 40% open, FCV actuator stem seized.

4. Components affected by the failure

- Water wall / evaporator tubes - primary risk of short-term overheating (DNB) and long-seam bulging.

- Steam drum internals - drum separators / cyclones subjected to swing conditions.
- Downcomers and bottom ring header - loss of circulation head.
- Superheater tubes - secondary overheating due to reduced steam flow after trip.
- Boiler feed pump - risk of cavitation and mechanical seal damage during low-flow operation.
- Feed control valve and its actuator.
- Drum level transmitters / condensate pots (measurement chain).

5. Initial hypotheses for the failure mechanism

- H1 - Feed-side hydraulic deficiency: BFP degradation or suction problem, unable to deliver rated flow at drum pressure.
- H2 - Feed path restriction / control failure: feed control valve stuck, strainer choked, or MOV throttled.
- H3 - Uncontrolled water loss from the circuit: inadvertent opening of CBD / EBD / low-point drain.
- H4 - Water wall or economiser tube leak dumping water into the furnace / gas path.
- H5 - False level indication: condensate pot leak or transmitter drift showing low level when the true level is normal.
- H6 - Sudden load swing (load throw-on) causing shrink and a transient level dip.

5.1 Failure modes observed

- Functional failure - loss of feed water inventory control.
- Component failure mode - feed control valve actuator seizure (stiction / stem galling).
- Component failure mode - CBD isolating valve left in partially open position (human/administrative failure mode).
- Latent failure mode - standby BFP auto-start circuit not functional (hidden failure revealed only on demand).

6. Which evidence supports which hypothesis

- The persistent feed-minus-steam negative mismatch supports H1, H2 and H3 together, and immediately weakens H5 and H6 (a false level or a shrink transient would not produce a real flow imbalance).
- Feed control valve saturated at 100% with no flow response supports H2 strongly and H1 weakly.
- Absence of furnace noise, absence of a drop in boiler water conductivity, and a stable flue gas temperature profile argue against H4.
- Rising boiler water conductivity supports H3 only partially (blow-down removes concentrated water, so a large CBD loss would normally reduce conductivity) - it indicates the dominant loss was not blow-down alone, pointing back to H2 as the main contributor with H3 as an aggravating factor.
- Failure of the standby BFP to start on auto is corroborating, not causal, evidence - it explains why the event escalated rather than why the level fell.
- Physical confirmation of the seized FCV stem and the part-open CBD valve closed the loop.

6.1 Most discriminating evidence

The most discriminating evidence is the combination of a saturated (100%) feed control valve demand with a flow that does not respond. It separates a control/actuation failure from every other cause: a pump problem would show low discharge pressure with the valve modulating, a tube leak would show furnace noise plus a conductivity drop, and an instrument fault would show no real flow imbalance at all.

7. How competing hypotheses were ruled out

- H1 (BFP degradation) ruled out - pump discharge pressure and motor current were within normal band for the flow delivered, and pump performance was verified against its head-flow curve after the event; the shortfall was downstream of the pump.

- H4 (tube leak) ruled out - no hissing from the furnace, no unstable draught, no drop in boiler water conductivity, no rise in flue gas moisture, and a subsequent hydraulic test at 1.25 times design pressure showed no leakage.
- H5 (false indication) ruled out - two independent transmitters plus the direct-reading gauge glass agreed within 2%, and the real flow imbalance was independently measurable.
- H6 (load swing) ruled out - TG load, steam flow and header pressure were flat throughout; a shrink transient would also self-correct in minutes rather than decline monotonically for 30 minutes.
- H3 (blow-down) retained but demoted to contributory - the part-open CBD accounted for only a fraction of the measured deficit and could not by itself have driven the level to the trip point at the observed rate.

8. Final failure mechanism determined

Loss of feed water regulating capability due to seizure of the feed control valve actuator stem, so that the valve could not open beyond its as-found position despite a 100% control demand. The resulting sustained feed deficit, aggravated by an inadvertently part-open continuous blow-down valve and by a latent failure of the standby BFP auto-start, produced a progressive fall in drum level towards the water-wall starvation regime.

9. Operational and maintenance context relevant to root cause

- The FCV had not been stroke-tested during the last two annual shutdowns; the plant maintenance strategy for the valve was run-to-failure by default.
- Feed water chemistry records showed repeated high-hardness excursions, consistent with deposition on the valve trim and stem.
- The CBD valve is manually operated in the field with no position feedback in DCS and no shift line-up checklist entry.
- The standby BFP auto-start had not been function-tested since commissioning; there is no periodic hidden-failure test in the PM schedule.
- The operator responded to the alarm rather than to the trend, so the deviation ran for approximately 25 minutes before it was acted on.

10. Final root cause

Physical root cause: stem seizure of the feed water control valve from deposit build-up and lack of exercise. Systemic root cause: absence of a preventive maintenance and functional-test regime covering (a) periodic stroking and calibration of the feed control valve, (b) proof-testing of the standby BFP auto-start (a hidden failure), and (c) field valve line-up verification. The failure mechanism (actuator seizure) is converted into a root cause by asking why the seizure was allowed to reach the point of demand: because no task existed to detect or prevent it.

11. Corrective actions to be taken

- Immediate: overhaul / replace the FCV actuator and trim; stroke-test and calibrate; restore the CBD valve to the closed position and lock it.
- Immediate: restore and function-test the standby BFP auto-start on discharge header pressure and on running-pump trip.
- Short term: verify feed water station line-up - FCV, its bypass, parallel line valve and all MOVs - and record positions in the shift log.
- Short term: introduce a DCS deviation alarm on (feed flow minus steam flow) sustained beyond a set band, so the operator is alerted by trend rather than by level alarm.
- Medium term: add quarterly stroking and annual overhaul of the FCV to the PM schedule; add six-monthly proof-testing of standby BFP auto-start.
- Medium term: enforce feed water and boiler water chemistry limits; investigate the hardness excursions at the DM plant / condensate polishing unit.

- Long term: provide position feedback for the CBD valve in DCS, and train operators on the low-drum-level actions sequence - attempt control with feed valve, open bypass / parallel line if stuck, close economiser inlet valve and run the pump on recirculation, and allow the boiler to trip on drum level low-low rather than force-running below safe level.

Case 2 - Drum Level High with carry-over into the superheater and turbine

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attemperator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. Drum internals comprise turbo-separators and secondary scrubber elements; drum level high alarm 68.5% and trip 75%.

2. Incident summary - what happened

Following a sudden reduction of TG load from 14 MW to 8 MW, drum pressure rose and the drum swelled. The three-element controller, whose steam flow signal had a sluggish transmitter, continued to admit feed water. Level rose past the high alarm to the high-high band. Main steam temperature fell sharply by about 45 degC in a few minutes, the turbine inlet temperature dropped and the turbine expansion trend turned negative. The boiler tripped on drum level high-high. Subsequent inspection showed salt deposits on the primary superheater and heavy fouling of the first turbine stages.

2.1 Timeline and onset

- T-0:00 - Steady 92% MCR, drum level 50%, MS temperature 490 degC.
- T+0:02 - TG load thrown off from 14 MW to 8 MW; steam flow drops rapidly, drum pressure rises.
- T+0:03 - Drum swell: level jumps to 62% within seconds; feed valve does not close proportionally.
- T+0:06 - Drum level High alarm at 68.5%; operator opens IBD.
- T+0:08 - Main steam temperature falls from 490 degC to 445 degC; superheater metal temperatures also fall - classic carry-over signature.
- T+0:10 - Level 75%, boiler trip on drum level high-high; boiler stop valve closed; turbine tripped by operator on falling steam temperature to protect against water induction.
- T+2:00 onwards - Boiler drains opened, unit cooled, inspection carried out.

3. Evidence, in order of observation

- Sharp fall in main steam temperature coincident with, not preceding, the level rise - the ordering distinguishes carry-over from a firing-side temperature loss.
- Simultaneous fall in superheater metal temperatures across all measured points.
- Steam purity / conductivity of saturated steam sample rising sharply.
- Negative turbine expansion trend and rising differential expansion.
- Feed flow remaining high while steam flow had already collapsed.
- Steam flow transmitter response lag observed on trend replay compared with turbine stop valve position.
- Post-inspection: salt and silica deposits on primary superheater tubes and turbine nozzle block; increase in specific steam consumption of the TG measured on restart.

4. Components affected by the failure

- Steam drum internals - separators / cyclones and scrubbers, subjected to overload and possible mechanical damage.
- Primary and platen superheater tubes - salt deposition, local overheating and long-term creep risk.

- Main steam piping and drains - risk of water slugging and hammer.
- Turbine HP nozzle block, first stages and blading - deposit fouling, thrust bearing risk from water induction.
- Main steam stop valve and ESV seats - erosion from wet steam.
- Feed water control loop - steam flow transmitter.

5. Initial hypotheses for the failure mechanism

- H1 - Drum level controller failure or mis-tuning, feeding water during a shrink-swell transient.
- H2 - Faulty steam flow signal to the three-element controller, so the controller could not see the load collapse.
- H3 - Feed control valve stuck open / passing.
- H4 - Genuine excess feeding by the operator on manual control.
- H5 - False high level from a plugged or leaking condensate pot / reference leg.
- H6 - Firing-side cause for the steam temperature fall (bed temperature collapse, excess attemperation), with the level rise being incidental.

5.1 Failure modes observed

- Functional failure - loss of drum level control during a load-rejection transient.
- Component failure mode - steam flow transmitter sluggish response (drift / impulse line partial blockage).
- Consequential failure mode - moisture and salt carry-over into the superheater and turbine.
- Damage mode - deposition-induced turbine efficiency loss and superheater fouling.

6. Which evidence supports which hypothesis

- The level rise beginning within seconds of load rejection supports H1 and H2 (swell-driven) and argues against H4 (a manual over-feed would not be so precisely time-correlated).
- The trend replay showing the steam flow signal lagging the actual turbine valve movement is direct support for H2.
- The fall in steam temperature occurring after the level rise, together with the fall in superheater metal temperature and rising steam conductivity, supports the carry-over mechanism and rules against H6.
- The feed control valve was seen to modulate and to close fully after the trip, which weakens H3.
- Two independent level measurements plus the gauge glass agreed, and the water level was physically confirmed after the trip, weakening H5.

6.1 Most discriminating evidence

The most discriminating evidence is the sequence and correlation: steam temperature and superheater metal temperature both fall only after drum level crosses the high band, while saturated steam conductivity rises at the same instant. A firing-side temperature loss would move steam temperature and metal temperature down without any change in steam purity and without preceding level movement.

7. How competing hypotheses were ruled out

- H1 (controller mis-tuning alone) ruled out as primary - the controller output was correct for the inputs it received; the fault was in the input signal, not the algorithm. Tuning remained a contributory weakness for shrink-swell handling.
- H3 (valve stuck open) ruled out - post-event stroking and seat leak test of the FCV were satisfactory, and the valve tracked demand during the event.
- H4 (operator over-feed) ruled out - the loop was in auto throughout; DCS mode logs confirm no manual intervention until the IBD was opened.
- H5 (false indication) ruled out - independent transmitters and gauge glass agreed, and the physical water level after trip matched the last indication.

- H6 (firing-side) ruled out - bed temperatures stayed within 850-890 degC, attemperator spray valve was closed, and fuel input had not changed; there was no independent reason for a 45 degC drop.

8. Final failure mechanism determined

Loss of drum level control during a load-rejection-induced swell, caused by a sluggish steam flow input to the three-element control scheme. With the steam flow signal lagging, the controller under-reacted to the collapse in demand and continued admitting feed water on top of the swell, driving the level into the drum internals and producing moisture and dissolved-solids carry-over into the superheater and turbine.

9. Operational and maintenance context relevant to root cause

- Impulse lines and transmitters for steam flow were last calibrated more than 24 months before the event; there is no periodic response/step check, only a static calibration.
- The three-element controller had never been re-tuned for the sharp load-rejection profile that the extraction TG can impose.
- The unit had experienced repeated grid-side load throw-off events in the preceding quarter, i.e. the transient was foreseeable and recurrent.
- Boiler water TDS was being run near the upper control limit to save blow-down, which increased the severity of the carry-over once it started.
- Operating instruction for drum level high did not clearly require closing the boiler stop valve immediately once final steam temperature starts dropping.

10. Final root cause

Physical root cause: sluggish / drifted steam flow measurement disabling the feed-forward action of the three-element drum level controller during a load rejection. Contributing root cause: boiler water solids maintained near the upper limit, which turned a level excursion into a damaging carry-over. Systemic root cause: the instrumentation PM programme verified static calibration only and never verified dynamic response, and the control scheme was never validated against the plant's known load-rejection transient.

11. Corrective actions to be taken

- Immediate: clean and prove the steam flow impulse lines, replace / recalibrate the transmitter, and verify dynamic response with a step check.
- Immediate: chemically clean or inspect the superheater as required and borescope the turbine first stages; record TG heat rate before and after.
- Short term: re-tune the three-element level controller specifically for the load-rejection profile, including feed-forward gain and swell compensation.
- Short term: restore boiler water TDS / silica to the middle of the control band and increase IBD/CBD accordingly.
- Medium term: add a continuous saturated-steam conductivity monitor with alarm, as an early carry-over detector independent of level.
- Medium term: revise the drum-level-high SOP - operate IBD, verify FCV position in DCS and field, close feed water station MOVs as required, watch PSH outlet steam temperature and close the boiler stop valve immediately if final steam temperature starts falling, and trip the turbine prime mover on falling steam temperature.
- Long term: install a drum-level-high-high interlock check in the annual protection-testing schedule; add water-induction protection review for the TG per the turbine OEM requirement.

Case 3 - Main Steam Temperature High

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attemperator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. Spray attemperation between LTSH and PSH, HP heater in the feed cycle, MS temperature design 495 degC with alarm at 550 degC.

2. Incident summary - what happened

After the HP heater was taken out of service for a tube leak, feed water temperature to the economiser dropped by about 30 degC. To hold steam pressure, the firing rate was increased. Main steam temperature climbed from 490 degC to 545 degC. The attemperator spray valve went fully open without arresting the rise. Superheater tube metal temperatures approached their alarm limits and positive turbine expansion increased. Load was reduced and the unit stabilised without a trip.

2.1 Timeline and onset

- T-0:00 - HP heater isolated for a leak; economiser inlet FW temperature falls from 165 degC to 135 degC.
- T+0:05 - Drum pressure sags; firing rate increased through coal feeders on all compartments.
- T+0:15 - MS temperature rising steadily, approx 3 degC per minute; spray flow increasing.
- T+0:25 - Spray control valve at 100%, MS temperature 535 degC and still rising.
- T+0:33 - MS temperature 545 degC, alarm at 550 degC approaching; SH metal temperatures high; positive turbine expansion rising.
- T+0:36 - Load reduced by 15%, upper-compartment coal feeding cut back, excess air reduced; temperature turns down.
- T+1:00 - MS temperature restored to 500 degC; spray valve back to 40%.

3. Evidence, in order of observation

- Loss of HP heater and a measured 30 degC drop in economiser inlet feed water temperature - the initiating change.
- Increase in firing rate at essentially constant load to hold pressure.
- Rising main steam temperature with attemperator spray demand saturating at 100%.
- Rise in superheater tube metal temperatures across the SH bank.
- Increase in positive turbine expansion and creep in tube-metal / steam-piping temperatures.
- Flue gas temperature after superheater rising, indicating a shifted heat-absorption profile.
- No evidence of soot / slag build-up on the water walls at this stage (bed temperature and evaporator absorption normal).

4. Components affected by the failure

- Superheater tubes - primary risk: short-term and long-term overheating, oxide scale growth, creep damage.
- Superheater outlet header and main steam piping - creep and thermal expansion stress.

- Attemperator / desuperheater and its spray control valve - risk of erosion and of thermal shock at the liner.
- Turbine HP casing, rotor and blading - overheating and differential expansion.
- Main steam stop valve and ESV - high-temperature service margin reduced.
- Condenser - increased heat load as a downstream effect.

5. Initial hypotheses for the failure mechanism

- H1 - Loss of HP heater causing low feed water temperature and a consequent increase in firing rate and superheat.
- H2 - Inadequate spray / attemperator capacity or a failed spray control valve.
- H3 - Excess air (high O₂) shifting heat pick-up towards the convective superheater.
- H4 - Soot lodging on water walls reducing evaporator absorption and raising gas temperature entering the SH.
- H5 - Delayed combustion / over-feeding of the upper compartments carrying the flame front upward.
- H6 - Faulty steam temperature indication.

5.1 Failure modes observed

- Functional failure - loss of final steam temperature control.
- Failure mode - superheater short-term overheating (metal temperature excursion, accelerated steam-side oxide growth).
- Failure mode - attemperation capacity exhaustion at reduced feed water temperature.
- Latent damage mode - creep life consumption in SH tubes, outlet header and MS piping.

6. Which evidence supports which hypothesis

- The tight time correlation between HP heater isolation, the feed water temperature drop and the onset of the temperature rise supports H1 above all others.
- The spray valve going to 100% without arresting the rise supports both H1 (demand exceeded capacity) and H2 (capacity failure) - these are distinguished by checking whether spray flow actually increased with valve opening.
- Measured spray flow rising in proportion to valve opening confirms the valve was working, so the deficiency was demand-side, i.e. H1.
- O₂ trending within the normal band and unchanged FD/SA damper positions weaken H3.
- Normal water-wall heat absorption, normal bed temperature and no rise in economiser inlet gas temperature weaken H4.
- Compartment-wise feeder trends show a uniform increase, not a top-heavy bias, weakening H5.

6.1 Most discriminating evidence

The most discriminating evidence is the measured spray flow increasing in step with the 100% valve opening while the steam temperature still rises. This proves the attemperator hardware is healthy and that the disturbance is an increased heat-input-to-steam-flow ratio caused by the low feed water temperature and the raised firing rate, not a control-element failure.

7. How competing hypotheses were ruled out

- H2 (spray system failure) ruled out - spray control valve stroked correctly and spray flow measurement increased with demand; the block valve was fully open and the spray water differential pressure was adequate.
- H3 (excess air) ruled out - O₂ at economiser outlet stayed in the 3-4% band and no damper had been repositioned.
- H4 (water-wall soot lodging) ruled out - evaporator absorption and furnace exit gas temperature were normal, and boiler cleanliness had been verified two days earlier.

- H5 (delayed combustion) ruled out - all six compartment bed temperatures moved together within a narrow band; no evidence of upward flame shift or of unburnt carry-over.
- H6 (faulty indication) ruled out - two independent MS thermocouples plus the TG inlet temperature and the rising SH metal temperatures agreed.

8. Final failure mechanism determined

Superheater outlet temperature excursion driven by an increased heat-release-to-steam-flow ratio. Removing the HP heater lowered feed water temperature, which increased the fuel required to hold pressure; the additional firing raised furnace exit and convective-pass gas temperature, increasing superheater absorption beyond the design attemperation capacity, so final steam temperature rose despite a fully open spray valve.

9. Operational and maintenance context relevant to root cause

- The HP heater had a known tube leak history and no standing operating instruction requiring an automatic load reduction when it is isolated.
- Operating philosophy prioritised holding steam pressure and load over holding steam temperature, so firing rate was increased rather than load reduced.
- The attemperator is single-stage and sized for design feed water temperature; the loss of the heater removes the design margin.
- SH metal temperature high alarms exist but the operators had no defined action limit tied to them.
- Coal being fired was higher in CV than design, so the same feeder speed delivered more heat.

10. Final root cause

Physical root cause: loss of the HP heater and the resulting low economiser inlet feed water temperature, compensated by increased firing at constant load, which exceeded the design attemperation capacity.

Systemic root cause: absence of an operating rule linking HP heater unavailability to a mandatory load reduction and to a revised steam temperature control strategy, plus a maintenance backlog on the HP heater itself.

11. Corrective actions to be taken

- Immediate: reduce load / firing rate, cut back feeding to the upper compartments, and reduce excess air where possible to bring MS temperature back within limits.
- Immediate: verify spray control valve, block valve and spray water source availability.
- Short term: repair the HP heater tube leak and return it to service; where HP heaters are available, keep them in service and, if not, reduce load as a standing rule.
- Short term: define and post action limits on SH tube metal temperature, with mandatory load reduction beyond a stated value.
- Medium term: review attemperator sizing for HP-heater-out operation; evaluate a second-stage spray or a burner tilt / upper-level fuel-cut strategy.
- Medium term: correct fuel-CV compensation in the combustion control so a richer coal does not silently raise heat input.
- Long term: introduce SH oxide-scale / creep-life monitoring at overhauls for tubes that have seen repeated high-temperature excursions.

Case 4 - Main Steam Temperature Low with water hammer risk

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attemperator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. Attemperator with a by-pass line, drum separators of turbo type, main steam line fitted with line drains and a bypass to PRDS.

2. Incident summary - what happened

Following a sharp increase in TG load, the boiler firing lagged, air flow was low and drum level rose. Main steam temperature fell from 492 degC to 428 degC in about eight minutes. Steam sample conductivity rose. A knocking sound was reported from the main steam line near the boiler stop valve. Turbine differential expansion moved negative. The operator opened the main steam line drains, reduced load and cut spray to zero; temperature recovered without turbine damage.

2.1 Timeline and onset

- T-0:00 - Stable at 88% MCR, MS temperature 492 degC.
- T+0:01 - TG load increased sharply from 11 MW to 14.5 MW; steam flow rises faster than firing rate responds.
- T+0:03 - Drum level rises on swell; attemperator spray valve found passing, spray flow still 3 TPH though demand is zero.
- T+0:05 - MS temperature falling; superheater metal temperature falling with it.
- T+0:08 - MS temperature 428 degC; saturated steam conductivity high; knocking sound reported at the MS line.
- T+0:09 - MS line drains opened, load reduced, spray isolated at the block valve.
- T+0:20 - MS temperature recovering; drains closed after clear steam observed.

3. Evidence, in order of observation

- Sharp load increase preceding the temperature fall - a load / firing mismatch signature.
- Drum level rise (swell) at the same time, followed by a slow return.
- Attemperator spray flow indicated with zero spray demand - a passing spray control valve.
- Main steam temperature and superheater metal temperature falling together.
- Saturated steam and main steam conductivity rising - moisture carry-over.
- Audible knocking / hammering in the main steam line.
- Negative turbine differential expansion trend.
- Air flow found to be at the low end of the required band for the new firing rate; O₂ dipped below 2%.

4. Components affected by the failure

- Superheater tubes - risk of thermal shock and of water-logging in individual elements.
- Main steam piping, hangers and supports - water hammer loading.
- Boiler stop valve and turbine ESV / control valves - erosion and possible seat damage from wet steam.
- Turbine LP-end and last-stage blades - water droplet erosion risk from wet, low-temperature steam.
- Turbine casing and rotor - negative expansion / rubbing risk.

- Attemperator spray control valve and its block valve.
- Drum separators - overloaded during the swell.

5. Initial hypotheses for the failure mechanism

- H1 - Sharp load increase with lagging firing rate and inadequate air flow (combustion deficit).
- H2 - Passing / stuck attemperator spray valve or a leaking attemperator by-pass, over-spraying the steam.
- H3 - High drum level causing moisture carry-over into the superheater.
- H4 - Faulty drum separators allowing carry-over even at normal level.
- H5 - Too high feed water temperature or excessive economiser absorption reducing gas temperature to the SH.
- H6 - Faulty steam temperature indication.

5.1 Failure modes observed

- Functional failure - loss of final steam temperature within the required band.
- Failure mode - attemperator spray control valve seat leakage (passing in the closed position).
- Consequential failure mode - wet steam / moisture carry-over and water hammer in the main steam line.
- Risk mode - turbine water induction and negative differential expansion.

6. Which evidence supports which hypothesis

- The temperature fall starting immediately after the load step supports H1.
- Measured spray flow with zero spray demand is direct, unambiguous support for H2.
- The concurrent level rise and rising steam conductivity support H3 as a contributor.
- Superheater metal temperature falling together with steam temperature is consistent with both H1 and H2 and excludes H6.
- The knocking sound with negative differential expansion confirms genuine wetness in the line, not an indication artefact.
- Economiser outlet FW temperature and gas temperatures were normal, giving no support to H5.

6.1 Most discriminating evidence

The most discriminating evidence is spray flow being registered while the spray demand is zero and the controller output is closed. That single observation identifies a passing spray valve, which no other hypothesis explains, and it is the difference between a temporary load-transient dip and a sustained low-temperature condition.

7. How competing hypotheses were ruled out

- H4 (faulty separators) ruled out at this stage - carry-over occurred only when the level was high; at normal level, steam purity was within limits both before and after the event. Separator health was nevertheless scheduled for inspection at the next opportunity.
- H5 (excess economiser absorption / high FW temperature) ruled out - economiser inlet and outlet temperatures and the gas-side profile were normal for the load.
- H6 (faulty indication) ruled out - independent thermocouples, TG inlet temperature, falling SH metal temperature and the physical knocking all agreed.
- H1 (load / firing mismatch) retained as the trigger but not the root cause - firing and air flow did catch up within minutes, whereas the low-temperature condition persisted until the spray block valve was closed.
- H3 (high drum level) retained as a contributor - it aggravated wetness but the level itself returned to normal before the temperature did.

8. Final failure mechanism determined

Sustained over-attenuation caused by seat leakage of the attenuator spray control valve, unmasked by a sharp load increase in which the firing rate and air flow lagged the steam demand and the drum swelled. The combination of genuinely reduced superheat and injected spray water plus moisture carry-over produced wet, low-temperature steam and condensate accumulation in the main steam line, giving rise to water hammer and negative turbine differential expansion.

9. Operational and maintenance context relevant to root cause

- The spray control valve had been in service for several years without seat overhaul; passing had been noticed earlier by field operators but not raised as a defect.
- The plant permits rapid TG load ramps without a matching boiler ramp-rate limit; there is no rate-of-change restriction in the operating instruction.
- Air flow was being run lean to reduce fan power, leaving little margin during a load pick-up.
- Main steam line drains are manually operated and are not routinely opened on a low steam temperature excursion.
- The alarm limit for low main steam temperature was set well below the value at which turbine protection concerns start.

10. Final root cause

Physical root cause: attenuator spray control valve seat leakage allowing uncontrolled spray water into the steam, compounded by a swell-driven carry-over. Systemic root cause: no periodic seat-tightness / passing test for the spray valve, no ramp-rate limit protecting the boiler from turbine load steps, and a defect-reporting culture in which a known passing valve was not raised as a work request.

11. Corrective actions to be taken

- Immediate: isolate the attenuator at the block valve, open main steam line drains, reduce load, and check drum level; if the rate of temperature drop is high, isolate the boiler and operate start-up as required.
- Immediate: verify that the turbine has not suffered water induction - check differential expansion, vibration and thrust bearing trends before reloading.
- Short term: overhaul or replace the spray control valve and its by-pass; carry out a seat-leak test.
- Short term: correct air flow control so that combustion air leads the fuel increase; restore O₂ to the design band.
- Medium term: impose and interlock a boiler-friendly TG load ramp rate; add a low-MS-temperature alarm with a defined operator action limit.
- Medium term: add periodic passing checks for all spray, drain and by-pass valves in the steam circuit.
- Long term: motorise the main steam line drains and interlock them with a low steam temperature signal; inspect drum separators at the next shutdown.

Case 5 - Main Steam Pressure High on sudden load rejection

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attemperator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. Boiler safety valves set at 72 and 74 kg/cm²(g), electromatic start-up vent valve, PRDS dumping station to the process header.

2. Incident summary - what happened

A grid disturbance caused the TG to trip on reverse power. Steam flow collapsed from 62 TPH to near zero within seconds while six compartments were burning at full feed. Drum pressure rose rapidly; the start-up vent was opened but the dumping (PRDS) control valve failed to open on demand. Both drum safety valves lifted at 72 and 74 kg/cm²(g). The boiler then tripped on drum level high following the pressure-driven shrink-swell disturbance. No mechanical damage occurred, but the safety valve seats subsequently passed.

2.1 Timeline and onset

- T-0:00 - 92% MCR, drum pressure 66 kg/cm²(g), MS flow 62 TPH.
- T+0:00:05 - TG trips on reverse power; ESV closes; steam flow falls to near zero.
- T+0:00:20 - Drum pressure rising rapidly at over 1 kg/cm² per 10 s; drum level swings violently.
- T+0:00:40 - Operator opens start-up vent valve; PRDS dumping control valve receives demand but does not open.
- T+0:01:10 - First drum safety valve lifts at 72 kg/cm²(g); second lifts at 74 kg/cm²(g).
- T+0:01:30 - Coal feeders cut back / DCFs stopped, wind box dampers throttled; bed temperature starts to fall.
- T+0:02:30 - Drum level high-high; boiler trips.
- T+0:06:00 - Pressure decaying; safety valves reseal but subsequently observed to pass.

3. Evidence, in order of observation

- TG trip signal and ESV closure recorded first - the initiating event.
- Collapse of main steam flow with fuel input unchanged.
- Rapid rise in drum pressure with a matching disturbance in drum level (shrink then swell).
- PRDS dumping control valve demand present in DCS with no corresponding valve position feedback movement.
- Both safety valves lifted at their set pressures - the protection functioned.
- Coal feeder speed remained at the pre-trip setting for approximately 70 s after the TG trip - a delayed fuel cutback.
- Post-event: safety valve seats found passing; drum level high-high trip recorded.

4. Components affected by the failure

- Drum and pressure parts - subjected to a rapid pressure transient and to the mechanical shock of safety valve lift.
- Drum safety valves - seat damage and subsequent passing (a common consequential failure).
- Superheater safety valve and its discharge piping - reaction loading.

- Drum internals - severe shrink-swell loading.
- PRDS / dumping control valve and its actuator.
- Start-up vent valve and its silencer.
- Feed water control system - large transient disturbance.

5. Initial hypotheses for the failure mechanism

- H1 - Sudden drop in steam flow / load rejection with fuel input not reduced quickly enough.
- H2 - Failure of the pressure relief path (PRDS dumping control valve) to open on demand.
- H3 - Uncontrolled fuel entry - coal feeder speed control fault or feeders not responding to cutback.
- H4 - Pressure control loop failure or wrong set point.
- H5 - Faulty pressure indication causing an unnecessary intervention.
- H6 - Safety valves set too high / not lifting at the correct pressure.

5.1 Failure modes observed

- Functional failure - loss of steam pressure control on load rejection.
- Component failure mode - PRDS dumping control valve fail-to-open on demand (hidden failure).
- Human/procedural failure mode - delayed fuel cutback after turbine trip.
- Consequential failure mode - safety valve seat damage and passing after lift.
- Consequential failure mode - drum level high-high trip from the pressure-induced swell.

6. Which evidence supports which hypothesis

- The TG trip preceding everything else and the unchanged fuel input support H1 as the trigger.
- Valve demand present with no position feedback change is direct support for H2.
- The 70 s lag before feeder cutback supports the procedural component of H3, but the feeders did respond correctly once commanded, which rules out a feeder control fault.
- Pressure controller output was correct and the set point was verified as normal, so H4 is unsupported.
- Both safety valves lifted at their stamped set pressures, which both confirms correct protection and refutes H6.
- Independent pressure transmitters and the local gauge agreed, refuting H5.

6.1 Most discriminating evidence

The most discriminating evidence is the PRDS dumping valve demand-versus-position mismatch. The load rejection alone would have been absorbed within the safety-valve margin had the dumping path opened; the fact that the valve was commanded and did not move is what converted a routine transient into a safety-valve lift.

7. How competing hypotheses were ruled out

- H3 (feeder control fault) ruled out - feeders reduced speed correctly the moment the command was given, and the drives showed no fault codes; the delay was in the human/logic response, not the equipment.
- H4 (pressure loop fault) ruled out - controller output, set point and tuning parameters were verified unchanged and correct.
- H5 (faulty indication) ruled out - three independent pressure measurements plus the physical lift of the safety valves at their stamped settings agreed.
- H6 (safety valve setting) ruled out - both valves lifted within tolerance of their set pressures; their subsequent passing is a consequence of the lift, not a cause of the event.
- H1 (load rejection) retained as the initiating event but not the root cause - load rejections are a foreseeable, designed-for condition; the design intent is that the dumping path and a fast fuel cutback absorb them.

8. Final failure mechanism determined

Over-pressurisation of the boiler following a turbine trip because the designed pressure relief route was unavailable and the fuel cutback was delayed. With steam flow at zero and six compartments still at full feed, the stored heat in the bed and the continuing heat release drove drum pressure up to the safety valve settings, causing both valves to lift and subsequently to pass.

9. Operational and maintenance context relevant to root cause

- The PRDS dumping control valve is exercised only during annual shutdown; it had not been stroked in service for over a year - a classic hidden failure.
- There is no automatic fuel-runback interlock on turbine trip; the coal feeder cutback is entirely manual.
- Grid disturbances causing TG trips had occurred three times in the preceding six months.
- The AFBC bed has very high thermal inertia, so cutting feeders alone does not remove heat release quickly - the operating procedure did not address bed slumping or damper throttling as a heat-removal action.
- Safety valve overhaul interval had been extended beyond the OEM recommendation.

10. Final root cause

Physical root cause: fail-to-open of the PRDS dumping control valve on demand, so the designed pressure relief route was not available during a foreseeable turbine trip. Systemic root causes: (a) no periodic in-service proof test of the dumping valve, allowing a hidden failure to accumulate, and (b) no automated fuel runback / bed-slumping interlock on turbine trip, leaving the response dependent on operator reaction time against the thermal inertia of the fluidised bed.

11. Corrective actions to be taken

- Immediate: overhaul the PRDS dumping control valve and actuator; prove it opens on demand from DCS.
- Immediate: lap or replace the drum safety valve seats and re-set / re-test the valves; passing safety valves are both a loss and a hazard.
- Short term: open the start-up vent valve promptly on any turbine trip, control fuel input and drum level, and check coal feeder feed rate as the standing first actions.
- Short term: brief operators that if the turbine has tripped first, the boiler should be allowed to trip on its own protection rather than being force-held, accepting that safety valves may lift.
- Medium term: implement an automatic fuel runback and compartment-slumping sequence interlocked to the turbine trip signal.
- Medium term: introduce monthly / quarterly partial-stroke testing of the dumping and start-up vent valves.
- Long term: restore the safety valve overhaul and pop-testing interval to the OEM / statutory requirement, and review whether the dumping capacity is adequate for a full load rejection.

Case 6 - Boiler outlet steam pressure too low (combustion-side deficit)

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attemperator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. Six independent bed compartments each with its own DCF and wind box damper; coal received through a common bunker with hoppers feeding each DCF.

2. Incident summary - what happened

During monsoon, wet fines-heavy coal caused hopper rat-holing on two compartments. Coal feed to those compartments became intermittent, their bed temperatures fell towards the low limit and boiler outlet steam pressure drifted down from 66 to 58 kg/cm²(g) at constant load. A DCF shear pin failed on a third compartment, further reducing feed. Frequency began dropping as the TG could not hold load. Burner support was taken and load was reduced until the hoppers were cleared.

2.1 Timeline and onset

- T-0:00 - 90% MCR, drum pressure 66 kg/cm²(g), all six bed temperatures 860-885 degC.
- T+0:20 - Compartments 2 and 5 bed temperature falling; feeder speed at maximum with no proportional rise in bed temperature.
- T+0:35 - Steam pressure down to 63 kg/cm²(g); PA header pressure normal.
- T+0:50 - DCF-4 chain stops - shear pin failure; compartment 4 bed temperature falls rapidly.
- T+1:05 - Steam pressure 58 kg/cm²(g); TG frequency dipping; O/L steam temperature also falling.
- T+1:10 - Start-up oil burners taken into service for support; load reduced by 20%.
- T+1:40 - Hoppers poked and cleared, shear pin replaced; feed restored; pressure recovering.

3. Evidence, in order of observation

- Falling boiler outlet steam pressure at constant load with no increase in steam demand.
- Bed temperatures on specific compartments falling while others remain normal - the pattern localises the fault to fuel feed, not to air or to heat transfer.
- Feeder speed at maximum with no corresponding rise in bed temperature - feed demand saturated without response.
- DCF-4 chain rotation stopped, shear pin found sheared.
- Coal moisture and fines content in the shift sample well above the design values.
- PA header pressure and air flow normal - excludes an air-side deficit.
- No abnormal furnace sound and no rise in feed water consumption - excludes a tube leak.
- Boiler safety valves and PRDS valves confirmed not passing.

4. Components affected by the failure

- Drag chain feeders - shear pin, chain, gearbox.
- Coal bunkers, hoppers and chutes - hang-up / rat-holing.
- Bed material and distributor plate - risk of defluidisation and of clinker at the low-temperature end if unburnt coal accumulates and then ignites.
- Bed coils / evaporator surface in the affected compartments - loss of duty.

- TG - operating below rated inlet pressure, with efficiency loss and frequency risk.
- Start-up oil burners - unplanned service.

5. Initial hypotheses for the failure mechanism

- H1 - Fuel feed interruption: hopper jamming, wet fines, or DCF mechanical failure.
- H2 - Insufficient combustion air or PA flow, preventing complete combustion.
- H3 - Excessive bed height causing poor fluidisation and reduced heat release.
- H4 - Passing or malfunctioning of boiler safety valves, PRDS control valve or dumping control valve, leaking steam away.
- H5 - A pressure-part leak (water wall or superheater) draining the circuit.
- H6 - Genuine load increase or a fall in feed water temperature increasing the heat requirement.
- H7 - Faulty pressure indication.

5.1 Failure modes observed

- Functional failure - inability to maintain rated outlet steam pressure.
- Component failure mode - DCF shear pin failure (overload protection acting on a jam).
- Material-handling failure mode - coal hang-up / rat-holing due to high moisture and fines.
- Consequential failure mode - localised bed temperature collapse with risk of defluidisation and of unburnt fuel accumulation.

6. Which evidence supports which hypothesis

- Compartment-selective bed temperature fall is the key pattern supporting H1 - an air-side or heat-transfer problem would affect the bed more uniformly.
- Feeder speed saturated with no response supports H1 and refutes any hypothesis in which fuel is reaching the bed.
- Normal PA header pressure, normal SA flow and normal O₂ refute H2.
- Bed height / bed differential pressure within normal band refutes H3.
- No abnormal furnace noise, no rise in make-up water and stable drum level refute H5.
- Constant TG load and normal feed water temperature refute H6.
- The physically confirmed sheared pin on DCF-4 and the poked-out hopper material confirm the mechanism directly.

6.1 Most discriminating evidence

The most discriminating evidence is the compartment-selective pattern of bed temperature loss combined with saturated feeder speed. Air-side, heat-transfer and steam-leakage causes all act on the boiler as a whole; only a fuel-delivery failure produces a fall confined to particular compartments while their feeders run at maximum.

7. How competing hypotheses were ruled out

- H2 (air deficit) ruled out - PA header pressure, SA flow and economiser outlet O₂ were all within the normal band, and no fan or damper had changed position.
- H3 (excessive bed height) ruled out - bed differential pressure was normal on all compartments and no bed material draining was required.
- H4 (steam leaking away) ruled out - safety valve tail-pipe temperatures were ambient, the PRDS and dumping control valves showed closed position with no downstream temperature rise, and no vent was passing.
- H5 (tube leak) ruled out - no hissing, stable drum level, no abnormal make-up, unchanged flue gas temperature and moisture.
- H6 (increased demand) ruled out - TG load, extraction flow and feed water temperature were all steady on trend.

- H7 (faulty indication) ruled out - drum pressure, MS pressure and TG inlet pressure all fell together and the local gauge agreed.

8. Final failure mechanism determined

Loss of heat release in individual bed compartments due to interruption of coal delivery. Wet, fines-rich coal bridged in the hoppers so that the drag chain feeders ran without delivering their rated flow, and on one compartment the resulting jam sheared the DCF shear pin and stopped the chain entirely. The cumulative fuel deficit reduced total heat input below that required for the load, so drum and outlet steam pressure fell.

9. Operational and maintenance context relevant to root cause

- Coal was being received and stored uncovered during monsoon; total moisture was well above the design basis for the feeders.
- No bunker vibrators, air blasters or poke-holes were provided on the affected hoppers, and hopper geometry favoured rat-holing with fines-heavy coal.
- Coal sizing / screening at the crusher house was not controlling fines within specification.
- Shear pin material and rating had been changed by local procurement without an engineering review.
- DCF chain and gearbox preventive maintenance had been deferred; no condition monitoring existed on chain rotation other than a proximity switch.
- Operators had no compartment-wise fuel-flow indication, only feeder speed - so a feed loss was visible only indirectly through bed temperature.

10. Final root cause

Physical root cause: interruption of coal flow to the bed caused by hopper hang-up of wet, fines-rich coal and consequent DCF jamming and shear pin failure. Systemic root cause: fuel quality (moisture and size) was not being controlled to the design specification, and the fuel-handling system had neither the flow aids nor the monitoring needed to detect and prevent hang-up before it became a heat-input deficit.

11. Corrective actions to be taken

- Immediate: check bed temperatures and increase coal feeding accordingly; clear jamming of coal hoppers and PA lines; take burner support if required; verify DCF chain rotation and replace failed shear pins.
- Immediate: ensure air flow is sufficient and drain bed material slowly if bed height is excessive.
- Immediate: confirm no passing of boiler safety valves, dumping control valve or PRDS control valve, and check for abnormal furnace sound before concluding the cause is fuel-side.
- Short term: if pressure continues to fall and frequency drops, reduce load on the turbine to protect the machine and the grid interface.
- Medium term: cover coal storage, blend to control moisture, and enforce crusher-house sizing so fines stay within specification; increase feed water temperature where possible to reduce the heat requirement.
- Medium term: fit hopper flow aids (air blasters / vibrators / poke doors) and restore engineered shear pin specification under a change-control process.
- Long term: provide compartment-wise coal flow or feeder-torque monitoring and alarm on chain rotation failure, so a feed interruption is detected directly instead of through bed temperature.

Case 7 - Bed Temperature High leading to clinker formation

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attemperator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. Bed temperature alarm 880 degC, DCF trip on bed temperature high at 940 degC and DCF reset at 930 degC; ash recirculation system with recirculation feeder and a sand feeder for bed make-up.

2. Incident summary - what happened

A consignment of high-CV, low-ash imported coal was fired without adjusting feeder speed or bed inventory. Bed temperature on compartments 3 and 4 climbed past 880 degC. PA flow was on the low side because a PA line was partially choked, and the ash recirculation feeder had tripped unnoticed. Bed temperature reached 940 degC and the coal feeders tripped on high bed temperature. On inspection, hard clinker had formed on the distributor plate in compartment 4 and two screen tubes showed local overheating discolouration.

2.1 Timeline and onset

- T-0:00 - Change to high-CV, low-ash coal; no change made to feeder speed or bed height.
- T+0:40 - Bed temperatures rising slowly across all compartments; compartments 3 and 4 leading.
- T+1:10 - Ash recirculation feeder trips; no separate alarm annunciated; bed inventory begins to fall.
- T+1:30 - Bed temperature high alarm at 880 degC on compartment 4; PA flow to that compartment noted below normal.
- T+1:50 - Attempted correction: recirculation ash increased (feeder found tripped), sand feeder started, PA flow increased.
- T+2:05 - Compartment 4 bed temperature 940 degC; coal feeders trip on bed temperature high.
- T+2:20 - Load reduced; compartment slumped and cooled.
- Next shutdown - Hard clinker found on the distributor plate; nozzle caps eroded; screen tube discolouration.

3. Evidence, in order of observation

- Change of fuel to a higher CV, lower ash coal immediately before the excursion - the initiating change.
- Progressive bed temperature rise strongest on the compartments with the lowest PA flow.
- PA flow / SA flow on the low side relative to fuel heat input; O₂ at economiser outlet falling.
- Ash recirculation feeder in the tripped state with no annunciation - loss of the primary bed-temperature moderating mechanism.
- Falling bed differential pressure indicating reduced bed inventory / height.
- Bed temperature thermocouples on the affected compartment reading consistently, and cross-checked against neighbouring compartments.
- Physical evidence at shutdown: hard sintered clinker on the distributor plate, eroded nozzle caps, discoloured screen tubes.

4. Components affected by the failure

- Bed distributor plate and air nozzles - clinker adherence, nozzle blockage and erosion.

- In-bed evaporator coils - overheating, erosion and possible tube failure.
- Screen tubes and refractory - refractory failure risk and screen tube overheating.
- Bed material - sintering / agglomeration, loss of fluidisation.
- Coal feeders / DCF - tripped by protection.
- Ash recirculation feeder and sand feeder.
- PA lines and wind box - partial choking.

5. Initial hypotheses for the failure mechanism

- H1 - High CV and low ash coal raising heat release per unit of bed material.
- H2 - Low PA flow / low SA flow, reducing the convective removal of heat from the bed and the fluidisation quality.
- H3 - Sudden change in load causing a fuel-air mismatch.
- H4 - Faulty bed temperature thermocouple giving a spurious high reading.
- H5 - Ash recirculation system trouble - the recirculation feeder not returning cool ash to the bed.
- H6 - Faulty bed ash draining, allowing bed inventory and particle size distribution to drift.
- H7 - Low bed height reducing thermal inertia and heat-absorbing surface contact.

5.1 Failure modes observed

- Functional failure - loss of bed temperature control.
- Failure mode - bed material sintering / agglomeration and clinker formation.
- Failure mode - distributor nozzle blockage and erosion.
- Hidden failure mode - ash recirculation feeder trip without annunciation.
- Consequential failure mode - screen and in-bed tube overheating; refractory damage.

6. Which evidence supports which hypothesis

- The fuel change immediately preceding the rise, with no compensating change in feeder speed, supports H1.
- The strongest temperature rise occurring on the compartments with the lowest PA flow supports H2 and localises it.
- The tripped recirculation feeder discovered mid-event, together with the falling bed differential pressure, supports H5 and H7 jointly - recirculation ash is both a coolant and a bed-inventory source.
- Consistency between multiple thermocouples on the same compartment and the physically observed clinker refutes H4.
- Steady TG load throughout the two hours refutes H3.
- Bed ash drain operation records were normal, giving no support to H6 as an initiator, though the reduced inventory made the bed more sensitive.

6.1 Most discriminating evidence

The most discriminating evidence is the tripped ash recirculation feeder combined with the falling bed differential pressure. Fuel CV and PA flow explain a general upward drift, but the loss of recirculated cool ash removes the specific mechanism the design relies on to hold bed temperature down, and it explains why the rise became uncontrollable rather than merely elevated.

7. How competing hypotheses were ruled out

- H3 (load change) ruled out - TG load and steam flow were flat; the excursion developed over two hours with no demand change.
- H4 (faulty thermocouple) ruled out - multiple thermocouples on the affected compartment agreed, adjacent compartments showed a consistent gradient, and clinker was physically found at that location.

- H6 (faulty bed ash draining) ruled out as an initiator - drain operations were logged as normal and bed particle size distribution from the shift sample was within band.
- H1 (fuel CV) retained as a genuine contributor but not sufficient alone - the boiler had previously fired similar coal successfully when recirculation and PA flow were normal.
- H2 and H7 retained as contributors - low PA flow and low bed height reduced the margin, but the excursion only ran away once the recirculation path was lost.

8. Final failure mechanism determined

Bed temperature runaway to the sintering range caused by loss of the ash recirculation cooling and bed-inventory path, in combination with an uncompensated increase in fuel calorific value and a partially choked PA line reducing fluidising air and heat removal on the affected compartments. Once local bed temperature approached the ash softening temperature, particles sintered together, forming hard clinker on the distributor plate and destroying fluidisation locally.

9. Operational and maintenance context relevant to root cause

- No formal fuel-change procedure existed requiring recalculation of feeder speed, bed height and air flow when coal CV or ash content changes.
- The ash recirculation feeder had no dedicated trip annunciation in DCS - its status was visible only on a sub-page.
- PA line cleaning was carried out only during shutdown; there was no in-service check of compartment-wise PA flow distribution.
- The sand feeder had not been used for several months and bed make-up practice was informal.
- Ash fusion temperature of the received coal was not being tested on receipt, so the operating margin to sintering was unknown.
- The DCF high-bed-temperature trip at 940 degC is the last line of defence; there was no intermediate automatic action such as an automatic feeder cutback at 900 degC.

10. Final root cause

Physical root cause: local bed sintering caused by bed temperature approaching the ash softening temperature, driven by loss of ash recirculation cooling, uncompensated high-CV fuel and restricted primary air. Systemic root cause: fuel property changes were not managed by procedure (no ash fusion testing on receipt, no mandated combustion re-setting), and the ash recirculation system - which is safety-relevant for bed temperature - was neither alarmed nor monitored as a critical function.

11. Corrective actions to be taken

- Immediate: control bed temperature by recirculating ash through the recirculation feeder; start the sand feeder if bed height permits; increase PA flow and SA flow; reduce load by cutting coal feeder speed.
- Immediate: if bed temperature rises above the limit, allow the coal feeders to trip and, if temperature exceeds the safe limit, allow the boiler to trip to save the bed from clinkerisation.
- Immediate: verify the bed temperature thermocouples before acting on a single reading.
- Short term: clear the choked PA line and verify compartment-wise air distribution; remove clinker and inspect / replace eroded distributor nozzle caps at the first opportunity.
- Short term: add a dedicated alarm for ash recirculation feeder trip and for low bed differential pressure.
- Medium term: institute ash fusion temperature and proximate analysis testing on every coal consignment, with a documented combustion re-setting procedure for CV and ash changes.
- Medium term: implement an automatic coal feeder cutback at an intermediate bed temperature so that the 940 degC trip is not the first automatic action.
- Long term: review bed inventory management, sand feeder availability and in-service PA line cleaning provisions.

Case 8 - Bed Temperature Low with risk of bed slumping and unburnt accumulation

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attenuator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. DCF trips on bed temperature low at 320 degC; light-up oil burners available for support; recirculation ash feeder in service.

2. Incident summary - what happened

At part load, the PA flow was left at a value appropriate for full load while coal feeding had been reduced. The resulting high excess primary air over-cooled the bed. A low-CV, high-ash coal was in use. Bed temperature fell steadily across all compartments, steam flow reduced, superheater temperature dropped and furnace draught became unstable. A coal feeder then tripped, and the affected compartment fell rapidly towards the low trip limit with unburnt coal accumulating in the bed - a re-ignition hazard.

2.1 Timeline and onset

- T-0:00 - Load reduced from 92% to 55% MCR; coal feeder speeds reduced; PA fan flow not correspondingly reduced.
- T+0:15 - All bed temperatures drifting down, 870 degC to 800 degC; O₂ at economiser outlet rising above 6%.
- T+0:30 - Steam flow reducing further; superheater outlet temperature dropping; furnace draught fluctuating.
- T+0:45 - Bed temperature 720 degC and falling; unburnt coal visually observed in the bed drain sample.
- T+0:55 - DCF-2 trips on a mechanical fault; compartment 2 bed temperature falls rapidly.
- T+1:05 - Recirculation ash feeder speed found high, adding cold ash to an already cool bed.
- T+1:10 - PA flow reduced, recirculation feeder slowed, bed material supply stopped, oil burner support taken on compartment 2.
- T+1:35 - Bed temperatures recovering above 800 degC; combustion stable.

3. Evidence, in order of observation

- PA flow high relative to load - the ratio of primary air to coal well above the design value.
- Rising economiser outlet O₂ (above 6%), confirming excess air.
- Uniform bed temperature fall across all compartments - a global rather than a localised cause.
- Reduction in boiler steam flow and a drop in superheater steam temperature.
- Fluctuating furnace draught - unstable combustion.
- Unburnt coal visible in the bed drain sample.
- Recirculation ash feeder running at high speed, adding cool material.
- Coal analysis showing low CV and high ash content versus design.

4. Components affected by the failure

- Bed material and distributor plate - defluidisation and unburnt fuel accumulation risk.

- In-bed evaporator coils - loss of duty, thermal cycling.
- Furnace and freeboard - risk of secondary combustion / puff if accumulated unburnt fuel re-ignites.
- Superheater - reduced steam temperature, thermal cycling.
- Coal feeders / DCF - one unit tripped.
- Recirculation ash feeder.
- PA fan and wind box dampers - operating away from the design point.

5. Initial hypotheses for the failure mechanism

- H1 - High PA flow relative to load, or high SA flow, over-cooling the bed.
- H2 - Low CV / high ash coal reducing heat release per unit mass fed.
- H3 - Coal feed interruption or coal feeder trip.
- H4 - Over-feeding of cold bed material or excessive recirculation ash into the furnace.
- H5 - Faulty bed temperature thermocouples.
- H6 - Water tube or screen tube leak quenching the bed.
- H7 - Sudden load reduction with the air side not re-matched.

5.1 Failure modes observed

- Functional failure - loss of bed temperature within the combustion window.
- Failure mode - over-cooling by excess primary air (an operating-envelope failure, not a component failure).
- Failure mode - accumulation of unburnt fuel in the bed, creating a deflagration hazard on re-ignition.
- Component failure mode - DCF mechanical trip on one compartment.
- Risk mode - bed defluidisation / slumping.

6. Which evidence supports which hypothesis

- The uniform fall across all six compartments points to a global cause - H1, H2 or H7 - and away from H3 and H6, which would act locally.
- High PA-to-coal ratio with O₂ above 6% is direct support for H1 and H7 together.
- Low CV, high ash coal analysis supports H2 as a contributor that reduced the margin.
- The high recirculation feeder speed supports H4 as an aggravating factor discovered during the response.
- Multiple thermocouples in agreement plus the physically observed unburnt coal and the falling steam flow refute H5.
- The absence of hissing, of a drum level disturbance and of any make-up water increase refutes H6.
- The DCF-2 trip explains only the late, compartment-specific acceleration, not the preceding hour of uniform decline.

6.1 Most discriminating evidence

The most discriminating evidence is the PA-flow-to-coal-flow ratio together with the economiser outlet O₂. A fuel-quality or a feed-interruption cause would show falling bed temperature with normal or falling O₂; only excess air shows falling bed temperature with clearly rising O₂ across all compartments simultaneously.

7. How competing hypotheses were ruled out

- H3 (coal interruption) ruled out as the initiator - feeder speeds and hopper levels were normal for the first 45 minutes; the DCF-2 trip occurred late and affected only one compartment.
- H5 (thermocouple fault) ruled out - all compartments and multiple elements agreed, and the steam-side response (falling steam flow and superheat) independently confirmed reduced heat release.
- H6 (tube leak quenching) ruled out - stable drum level, no abnormal make-up, no hissing from the furnace, no drop in boiler water conductivity, no unusual flue gas moisture.

- H2 (low CV coal) retained as a contributor - it reduced heat release per tonne but the boiler had operated on similar coal at correct air ratios without a bed temperature problem.
- H4 (excess recirculation ash) retained as an aggravating factor found during the response, not as the initiator, because the decline had begun before the feeder speed became relevant.

8. Final failure mechanism determined

Bed over-cooling by excess primary air. When load was reduced, coal feed was cut but PA flow was not re-matched, so the mass of fluidising air per unit of fuel rose sharply and carried heat out of the bed faster than combustion could replace it. Low-CV, high-ash coal and excessive cold recirculation ash further reduced the heat available, driving bed temperature down towards the trip limit and allowing unburnt coal to accumulate in the bed.

9. Operational and maintenance context relevant to root cause

- There is no automatic PA-flow-to-coal-flow ratio control; the air side is set manually and is typically left at the full-load setting during part-load operation.
- Part-load operation had become frequent because of process steam demand variation, but no part-load operating envelope had been defined.
- Coal CV was below the design basis and was not being blended.
- The recirculation ash feeder had no interlock with bed temperature.
- Operators had no clear instruction on the unburnt-fuel accumulation hazard when bed temperature is low; the natural response of increasing coal feed to raise temperature can make the hazard worse.
- DCF-2 had an open maintenance defect for chain misalignment.

10. Final root cause

Physical root cause: bed heat balance driven negative by excess primary air at part load, compounded by low-CV coal and excessive cold recirculation ash. Systemic root cause: absence of automatic (or proceduralised) air-to-fuel ratio control across the load range and absence of a defined part-load operating envelope, so that a routine load reduction moved the bed outside its stable combustion window without any alarm or interlock detecting it.

11. Corrective actions to be taken

- Immediate: reduce PA fan flow and reduce SA flow if excessive; reduce the speed of the recirculation ash feeder; stop bed material supply if running.
- Immediate: check bed temperature thermocouples and check the coal feeder and coal supply; take oil burner support to hold the affected compartment above the ignition temperature.
- Immediate: check for any leakage sound from the furnace before concluding the cause is air-side.
- Caution: do not simply increase coal feed into a bed already below the ignition temperature - clear the accumulated unburnt fuel and restore temperature with burner support first, to avoid a deflagration on re-ignition.
- Short term: rectify the DCF-2 chain misalignment defect and clear the open work order backlog on the feeders.
- Medium term: implement PA/SA-to-fuel ratio control with an O₂ trim, and define a documented part-load operating envelope with air-flow set points per load band.
- Medium term: interlock the recirculation ash feeder to bed temperature so it slows automatically as bed temperature falls.
- Long term: introduce coal blending / CV control on receipt, and train operators specifically on low-bed-temperature hazards and the correct recovery sequence.

Case 9 - Furnace Draught High and furnace explosion hazard

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC Atmospheric Fluidised Bed Combustion (AFBC) bi-drum boiler, Indian OEM class (Thermax / Cethar / ISGEC type), 6-compartment bed with 6 Drag Chain Feeders (DCF), 2x50% ID, 2x50% FD(SA), 2x50% PA fans, tubular APH-PA and APH-SA, single-stage attemperator between LTSH and PSH. Feeding a 15 MW extraction-cum-condensing TG set in a captive co-generation plant.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, Indian washery-reject / imported blended coal, design bed temperature 850-900 degC, drum level normal 50% (alarm Lo 20% / trip 10%, alarm Hi 68.5% / trip 75%), furnace draught maintained at -5 to -20 mmwc (alarm +50 / trip +100 mmwc, alarm -100 / trip -150 mmwc), main steam temperature alarm 550 degC, bed temperature alarm 880 degC with DCF trip at 940 degC and low-limit DCF trip at 320 degC. Furnace pressure alarm high +50 mmwc, trip +100 mmwc; alarm low -100 mmwc, trip -150 mmwc; ID fans on inlet damper control, explosion doors provided on the second pass.

2. Incident summary - what happened

During a restart after a flame failure, the purge was cut short because the operator over-rode the purge permissive to save start-up time. Un-burnt oil and coal from the failed light-up attempt remained in the furnace and the hot gas path. When the burner was re-lit, the accumulated fuel ignited, producing a pressure pulse that lifted the explosion doors, spiked furnace pressure to well above +100 mmwc and tripped the boiler on furnace draught high. Ducting expansion joints in the second pass were found deformed.

2.1 Timeline and onset

- T-0:00 - Flame failure on the light-up oil burner; boiler trips on flame failure. Un-burnt oil is present in the furnace.
- T+0:04 - Restart initiated; purge permissive over-ridden; purge duration approximately 90 s instead of the required five minutes at 40% air flow.
- T+0:07 - Burner spark energised; ignition of accumulated fuel-air mixture.
- T+0:07:02 - Furnace pressure spikes far above the +100 mmwc trip; explosion doors lift.
- T+0:07:03 - Boiler trips on furnace draught high; ID fan inlet dampers slam.
- T+0:15 - Inspection: explosion doors open, expansion joint on second pass ducting deformed, no personnel injury.
- T+1:00 - Furnace inspected, oil-soaked deposits found on the furnace floor and lower water wall.

3. Evidence, in order of observation

- Preceding flame failure with the fuel shut-off valve confirmed only after the event - fuel had continued to enter briefly.
- Purge permissive over-ride recorded in the DCS event log with a purge duration far shorter than required.
- Furnace pressure record showing a step spike rather than a ramp - characteristic of a deflagration, not of a fan or damper control fault.
- Explosion doors lifted - direct physical evidence of an over-pressure event.
- Flue gas analysis after the event showing carbon monoxide, consistent with an earlier accumulation of un-burnt fuel.
- Oil-soaked deposits on the furnace floor and lower water wall.
- Deformed expansion joint on the second pass ducting.
- ID and FD fan control found in auto and functioning normally throughout - no fan control fault.

4. Components affected by the failure

- Furnace walls, water wall panels and buckstays - over-pressure loading.
- Ducting, expansion joints and enclosures - the weakest parts, first to deform or fail.
- Explosion doors / relief panels.
- APH and ESP - pressure pulse propagation and potential secondary combustion.
- ID and FD fans - reverse-flow and blade loading during the pulse.
- Light-up oil system - burner tip, fuel shut-off valve, atomising steam / air.
- Refractory in the bed and freeboard region.

5. Initial hypotheses for the failure mechanism

- H1 - Furnace explosion from ignition of accumulated un-burnt fuel due to inadequate purge.
- H2 - Faulty operation of ID / FD / PA / SA fan control causing a pressurisation transient.
- H3 - Un-controlled fuel entry through a leaking oil shut-off valve or through excessive coal feeding.
- H4 - Disturbed combustion / secondary combustion in the back-pass.
- H5 - Sudden loss of ID fan or of ID damper control, pressurising the furnace.
- H6 - Faulty furnace pressure transmitter.

5.1 Failure modes observed

- Failure mode - furnace deflagration (rapid over-pressure) on re-ignition of an accumulated fuel-air mixture.
- Procedural failure mode - defeat of the purge permissive interlock.
- Component failure mode - oil burner fuel shut-off valve leakage allowing fuel accumulation.
- Consequential failure mode - ducting expansion joint deformation; explosion door lift.

6. Which evidence supports which hypothesis

- The purge over-ride in the event log, occurring after a flame failure with fuel present, is the direct causal chain for H1.
- The step-shaped pressure spike, rather than a ramp, distinguishes a deflagration (H1) from any fan or damper control transient (H2, H5), which would produce a comparatively slow pressure ramp.
- Explosion door lift and expansion joint deformation confirm the magnitude and speed of the event, consistent only with H1.
- Oil-soaked deposits and post-event carbon monoxide confirm the accumulation of un-burnt fuel and support H3 as a contributing cause.
- Fan control being in auto and both ID fans running with normal current refutes H2 and H5.
- Physical explosion door lift refutes H6 conclusively - the pressure was real.

6.1 Most discriminating evidence

The most discriminating evidence is the shape of the furnace pressure trace. A deflagration produces a near-instantaneous step spike that lifts relief devices; a fan or damper control failure produces a ramp over seconds to tens of seconds. Combined with the logged purge over-ride, this fixes the mechanism unambiguously.

7. How competing hypotheses were ruled out

- H2 (fan control fault) ruled out - both ID fans and both FD fans were in auto with normal motor current and normal damper feedback before and after the spike; no control action preceded the event.
- H5 (loss of ID fan) ruled out - no ID fan trip is recorded; the furnace pressure trip followed the spike rather than preceding it.
- H4 (secondary combustion in the back-pass) considered and partly retained - carbon monoxide indicated un-burnt fuel had reached the back-pass, but the primary over-pressure originated in the furnace, evidenced by the location of the deposits and the door that lifted.

- H6 (faulty transmitter) ruled out - explosion doors lifted and hardware deformed; the over-pressure was physically real.
- H3 (fuel entry) retained as a contributing cause - the shut-off valve was subsequently found to pass, which explains why fuel continued to accumulate after the flame failure.

8. Final failure mechanism determined

Furnace explosion (deflagration) caused by ignition of an accumulated un-burnt fuel and air mixture in a confined space. Following flame failure, fuel continued to enter through a passing oil shut-off valve and settled in the furnace. The mandated purge, which exists precisely to remove such a mixture, was defeated. On re-ignition the accumulated mixture burnt almost instantaneously, generating a pressure pulse that exceeded the furnace draught high trip and the structural capacity of the weakest enclosure elements.

9. Operational and maintenance context relevant to root cause

- Commercial pressure to restore the unit quickly created an incentive to shorten start-up; the purge permissive could be over-ridden from the DCS without a second authorisation.
- The light-up oil burner shut-off valve had a known passing history recorded in the shift log but no work order raised.
- Flame scanner and photocell cleaning was not part of a scheduled routine, and repeated flame failures had been treated as a nuisance rather than as a defect.
- Furnace safety system and trip interlocks were tested annually only; the purge timer and permissive logic had not been proof-tested that year.
- Flue gas was not routinely analysed for carbon monoxide, so incomplete combustion was not being detected.
- Explosion doors had not been inspected for free operation within the maintenance cycle.

10. Final root cause

Physical root cause: ignition of an accumulated un-burnt fuel-air mixture in the furnace, made possible by a passing oil burner shut-off valve and by the omission of the mandatory purge. Systemic root cause: a safety-critical interlock (purge permissive) could be defeated by a single operator without authorisation or record-based review, within a management system that tolerated known passing fuel valves and repeated flame failures as routine.

11. Corrective actions to be taken

- Immediate: do not restart until the furnace has been purged with approximately 40% of full load air flow for at least five minutes; no short cut in purging is permissible.
- Immediate: prove the light-up oil shut-off valve tight; ensure the idle burner fuel shut-off valve is closed and leak-tested.
- Immediate: inspect and reset the explosion doors, and repair the deformed expansion joint before restarting.
- Short term: remove the ability to over-ride the purge permissive from the operator interface, or require a dual-key / engineer-level authorisation with automatic event logging and management review.
- Short term: adjust the fuel-air ratio and verify the furnace safety system and trip interlocks are functioning correctly.
- Medium term: introduce routine flue gas analysis including carbon monoxide, so that incomplete combustion is detected before it accumulates.
- Medium term: maintain proper furnace draught with a functioning auto trip interlock on high furnace pressure, and proof-test the interlock and purge timer at a defined frequency.
- Long term: address the recurring flame failures at source - flame scanner cleaning, atomisation quality, oil quality and burner tip condition - so that the restart-after-flame-failure scenario becomes rare.

Case 10 - Flame Failure on the light-up oil burner

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set. Light-up system: pressure-jet / steam-atomised LDO burners with high-energy ignition, UV flame scanner and photocell.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. Oil burners are used only for cold start-up and for bed support at very low bed temperature; the boiler trips on flame failure during the light-up phase.

2. Incident summary - what happened

During a cold start-up after a shutdown, the oil burner flame kept extinguishing within 30-60 s of establishment. Three light-up attempts failed. Steam pressure could not be raised and the bed did not reach ignition temperature. Investigation found dirty, water-contaminated LDO, a choked burner tip, and a fouled photocell reading a weak flame signal even when the flame was visually present.

2.1 Timeline and onset

- T-0:00 - Cold start-up commenced; furnace purged; bed fluidised at low PA flow.
- T+0:06 - First light-up attempt: flame established, then lost after approximately 45 s; boiler trips on flame failure.
- T+0:14 - Second attempt after a fresh purge; flame unstable, visible smoke; loss of flame at 30 s.
- T+0:25 - Third attempt: flame visually present at the peep hole but the scanner shows a weak signal; trip on flame failure.
- T+0:40 - Burner withdrawn; tip found partly choked, atomising steam pressure low, LDO sample shows water and sediment.
- T+1:10 - Photocell cleaned, gap on spark plug reset, tip cleaned, oil drained and fresh oil taken; successful light-up on the fourth attempt.

3. Evidence, in order of observation

- Repeated loss of flame shortly after establishment, on successive attempts - a repeatable, not random, pattern.
- Visible smoky flame and poor atomisation at the peep hole.
- Atomising steam / air pressure lower than the burner OEM requirement.
- Weak flame scanner signal even when a flame was visually confirmed - pointing to a detection problem in addition to a combustion problem.
- LDO day tank sample showing free water and sediment.
- Burner tip nozzle holes found partly choked on withdrawal.
- Spark plug found dirty with an incorrect electrode gap.
- Combustion air / blower discharge pressure lower than specified.

4. Components affected by the failure

- Oil burner assembly - tip, nozzle, atomiser and diffuser.
- Flame scanner and photocell.
- High-voltage ignition transformer and spark plug.
- LDO supply system - day tank, strainers, pump, heater.
- Combustion air blower for the burner.
- Furnace - risk of accumulated un-burnt oil and consequent explosion hazard.

- Superheater - thermal cycling from repeated failed starts.

5. Initial hypotheses for the failure mechanism

- H1 - Poor fuel quality: dirty oil, water contamination, wrong viscosity.
- H2 - Improper atomisation: choked tip, low atomising steam pressure, worn nozzle.
- H3 - Faulty flame scanner / photocell giving a false loss-of-flame signal.
- H4 - Failure of the combustion air blower or insufficient combustion air.
- H5 - High furnace pressure disturbing the flame front.
- H6 - Ignition system fault - spark plug gap, HT transformer, ignition timing.

5.1 Failure modes observed

- Functional failure - inability to establish and sustain a stable light-up flame.
- Component failure mode - burner tip fouling and nozzle wear (degraded atomisation).
- Component failure mode - flame detector fouling causing signal degradation.
- Latent hazard mode - accumulation of un-burnt oil in the furnace after each failed attempt.

6. Which evidence supports which hypothesis

- Water and sediment in the oil sample plus a visibly smoky flame support H1 and H2 together.
- Low atomising steam pressure and the physically choked tip are direct support for H2.
- A weak scanner signal in the presence of a visually confirmed flame supports H3, but as a co-existing fault rather than as the sole cause - the flame was genuinely being lost on the first two attempts.
- Low blower discharge pressure supports H4 as a contributor to poor mixing.
- Furnace pressure was steady and within the normal band throughout, giving no support to H5.
- The dirty spark plug with an incorrect gap supports H6 as a contributor to difficult establishment, though the flame did establish each time before being lost.

6.1 Most discriminating evidence

The most discriminating evidence is the combination of a visually confirmed flame with a weak scanner signal on the third attempt, set against a visually smoky, unstable flame on the first two. This separates the two co-existing faults: attempts one and two were genuine combustion failures from poor atomisation and contaminated fuel; attempt three was a detection failure from a fouled photocell.

7. How competing hypotheses were ruled out

- H5 (high furnace pressure) ruled out - furnace draught was held between -5 and -15 mmwc on trend throughout all three attempts.
- H4 (blower failure) ruled out as a primary cause - the blower was running and delivering air, though at a marginal pressure; it degraded mixing but did not by itself extinguish the flame.
- H3 (scanner) ruled out as the cause of attempts one and two - the flame was independently observed to go out, so those were real losses.
- H1 and H2 retained jointly - the fuel contamination caused the tip choking, and both were physically confirmed.
- H6 (ignition system) retained as a contributor - it delayed establishment but did not explain loss of an already-established flame.

8. Final failure mechanism determined

Loss of a stable light-up flame caused by degraded atomisation. Water and sediment in the LDO choked the burner nozzle holes and, together with low atomising steam pressure and marginal combustion air, produced coarse droplets and incomplete combustion so that the flame could not sustain itself. A separately developing fouling of the flame photocell added a false loss-of-flame indication on the final attempt.

9. Operational and maintenance context relevant to root cause

- LDO tank water draining was not part of the routine shift checklist and the tank had not been drained for months.
- Oil strainers and burner tips were cleaned only when a problem occurred, not on a purge-cycle basis as recommended by the burner supplier.
- Flame scanner / photocell cleaning had no defined frequency.
- The plant had a long gap between cold starts, so the light-up system was effectively a standby system with hidden failures.
- Each failed attempt left un-burnt oil in the furnace, and the purge-before-restart requirement was under time pressure - creating the precondition for the explosion scenario described in Case 9.

10. Final root cause

Physical root cause: degraded fuel atomisation from water- and sediment-contaminated LDO choking the burner tip, aggravated by low atomising steam pressure. Systemic root cause: the entire light-up system is a standby system whose components (tips, strainers, scanners, spark plugs, tank drains) were maintained reactively, so multiple hidden failures accumulated and appeared together on demand.

11. Corrective actions to be taken

- Immediate: purge the boiler before putting the burner back, and purge the burner as per the cycle time given by the supplier - this is mandatory after every failed attempt.
- Immediate: clean the burner tip and check for choked nozzle holes; check and restore atomising steam / air pressure.
- Immediate: drain water from the LDO day tank, replace strainers and take fresh oil; check oil temperature and viscosity.
- Immediate: check the flame scanner and clean the photocell; check the sparking circuit and the high-voltage transformer supply; clean the spark plug and set the correct gap.
- Immediate: check the air blower and maintain the correct air-fuel ratio; check the explosion doors and close them if opened.
- Medium term: schedule periodic cleaning of tips, strainers, scanners and spark plugs, and periodic tank water draining, based on operating hours rather than on failure.
- Long term: treat the light-up system as safety-critical standby equipment with a documented functional test before every planned shutdown, so that it is known to be available before it is needed.

Case 11 - Water wall / screen / evaporator tube failure

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. Water walls of membrane construction; screen tubes at the bed-to-freeboard transition; low-point drains at the bottom ring header.

2. Incident summary - what happened

Over a period of about eight hours, boiler feed water consumption crept up relative to steam flow and boiler water conductivity began to fall. A faint hissing was heard near the front water wall during a field round. Bed temperature dropped sharply in one compartment, furnace draught became unstable and the flame appeared fluctuating. Load reduction was started at once and the unit was tripped before the leak enlarged. A single water wall tube was found with a longitudinal split adjacent to a heavily eroded, thinned section near the bed region, together with internal deposits.

2.1 Timeline and onset

- T-0:00 - Normal operation; feed flow and steam flow matched.
- T+2:00 - Make-up water consumption rising slightly; boiler water conductivity trending down.
- T+4:30 - Feed flow now consistently higher than steam flow by 2-3 TPH; drum level control working harder.
- T+6:00 - Faint hissing heard on the field round near the front wall at bed level; furnace draught starting to fluctuate.
- T+7:15 - Bed temperature in the adjacent compartment drops sharply; flame fluctuating; unstable draught.
- T+7:40 - Load reduction commenced; second BFP started as drum level could not be held by one pump.
- T+8:10 - Unit tripped in a controlled manner before the leak escalated; boiler isolated and cooled.
- Inspection - Longitudinal split in one water wall tube adjacent to an eroded, wall-thinned region; internal deposits present.

3. Evidence, in order of observation

- Rising feed water flow for a given steam generation, and rising make-up water consumption - the earliest quantitative indicator.
- Falling boiler water conductivity - the circuit is being diluted, the signature that distinguishes a leak from a blow-down or feed-control fault.
- Hissing steam leakage noise audible in the furnace region.
- Unstable, fluctuating furnace draught.
- Sharp drop in bed temperature local to the leak.
- Fluctuating flame / unstable combustion.
- Drum level becoming difficult to maintain with one BFP.
- Physical evidence at inspection: eroded, thinned tube wall with a longitudinal split; internal waterside deposits.

4. Components affected by the failure

- Water wall / evaporator tubes - the failed component and its neighbours, which are exposed to steam-jet cutting from the leak.

- Screen tubes at the bed transition - high erosion exposure.
- Bottom ring header and low-point drains.
- Refractory around the bed and the tube penetrations - washing and spalling.
- Bed material and distributor plate - quenching and possible defluidisation.
- Boiler feed pumps - forced into higher duty and possible parallel running.
- Drum internals - disturbed circulation.

5. Initial hypotheses for the failure mechanism

- H1 - Starvation / overheating from low drum level causing DNB and short-term overheat failure.
- H2 - Fireside erosion of the tube by bed material at the bed-to-freeboard transition, thinning the wall until it burst.
- H3 - Waterside deposits (blocked tube, salt deposition) causing local overheating and pitting.
- H4 - Circulation impaired by an open low-point drain, starving the affected circuit.
- H5 - Original manufacturing / weld defect.
- H6 - Corrosion - caustic gouging or hydrogen damage from poor water chemistry.

5.1 Failure modes observed

- Failure mode - fireside erosion-driven wall thinning progressing to a pressure-boundary rupture.
- Failure mode - waterside deposit accumulation with local overheating (contributory).
- Consequential failure mode - steam-jet erosion of adjacent tubes.
- Consequential failure mode - bed quenching, unstable draught and loss of combustion stability.

6. Which evidence supports which hypothesis

- The falling boiler water conductivity together with a rising feed-to-steam ratio establishes a pressure-part leak and immediately excludes all non-leak explanations of the water imbalance.
- The tube split being adjacent to a visibly eroded, thinned section, at the bed transition where bed material velocity is highest, is direct support for H2.
- The longitudinal (fish-mouth-free) split with wall thinning rather than a thick-lipped bulge distinguishes erosion from short-term overheating, weakening H1.
- Drum level records showing no excursion below the low alarm in the preceding weeks further weakens H1.
- Internal deposits support H3 as a contributory factor but not as the location-determining cause, since the failure occurred where erosion was greatest, not where deposits were heaviest.
- The characteristic morphology of the fracture, with no gouging or intergranular cracking, weakens H6.
- Metallurgical examination and the absence of any weld at the failure location weaken H5.

6.1 Most discriminating evidence

The most discriminating evidence is the fracture morphology and the wall thickness profile. Erosion produces progressive external wall loss with a thin-lipped longitudinal split at the thinnest point; short-term overheating produces a thick-lipped fish-mouth rupture with bulging; corrosion produces gouged or grooved internal loss. Ultrasonic thickness mapping of the surrounding area, showing a broad thinned zone, confirmed erosion.

7. How competing hypotheses were ruled out

- H1 (starvation overheating) ruled out - no low drum level excursions in the operating history, no bulging, no thick-lipped rupture, and no oxide scale thickening indicative of prolonged overheating.
- H4 (open low-point drain) ruled out - all low-point drain valves were verified fully closed and their downstream lines were cold.
- H5 (manufacturing defect) ruled out - the failure was in parent material remote from any weld, and no defect indications were present in the adjacent material.

- H6 (corrosion) ruled out - water chemistry records were within limits for the period, and the failure surface showed no gouging, no pitting concentration and no hydrogen damage features.
- H3 (waterside deposits) retained as contributory - deposits raised metal temperature locally and reduced the residual strength of the already-thinned wall, but they did not determine the failure location.

8. Final failure mechanism determined

Fireside erosion of the water wall tube by circulating bed material in the high-velocity bed-to-freeboard transition zone, progressively reducing wall thickness until the remaining section could no longer contain the drum pressure, at which point a longitudinal split opened. Internal deposits contributed by raising local metal temperature and reducing allowable stress.

9. Operational and maintenance context relevant to root cause

- The boiler had been operated for extended periods at high excess air, increasing gas and particle velocities in the erosion-prone zone.
- Coal ash was more abrasive than design, with high quartz content; no ash abrasiveness testing was in place.
- Bed height had frequently been run above the recommended band, increasing particle carry-over into the transition zone.
- No periodic ultrasonic thickness survey of the erosion-prone zone was carried out - inspection was visual only.
- Erosion shields / wear pads in the bed region had been removed at an earlier overhaul and not reinstated.
- Water chemistry had been generally within limits, but boiler cleanliness (chemical cleaning) was overdue.

10. Final root cause

Physical root cause: loss of pressure-boundary wall thickness by fireside particle erosion in the bed transition zone. Systemic root cause: erosion was a known and predictable degradation mechanism for this boiler type, yet the maintenance strategy contained no thickness-monitoring task, the protective wear shields had been removed without engineering review, and the operating envelope (excess air, bed height, ash abrasiveness) was not controlled to limit erosion rate. The failure mechanism is thus converted to a root cause by asking why a foreseeable wear-out mechanism was neither measured nor mitigated.

11. Corrective actions to be taken

- Immediate: as soon as leakage is noticed, start reducing load and trip the boiler in a controlled manner - do not continue running, since the leak enlarges rapidly and damages neighbouring tubes.
- Immediate: listen in the region for steam leakage and try to locate the leakage through manholes before the boiler is depressurised.
- Immediate: check that all low-point drain valves are fully closed, and do not allow the drum level to go beyond the danger level; start another BFP if the level cannot be maintained by the running pump.
- Short term: replace the failed tube and ultrasonically map wall thickness over the whole erosion-prone zone; replace all tubes below the minimum allowable thickness, not only the failed one.
- Short term: reinstate erosion shields / wear pads in the bed and screen region.
- Medium term: introduce a routine every-shift furnace listening check for steam noise and every-shift verification of low-point drain closure.
- Medium term: control excess air and bed height to the design band; test coal ash abrasiveness on receipt.
- Long term: schedule chemical cleaning to remove waterside deposits, and add periodic UT thickness surveys of water wall, screen and evaporator tubes into the shutdown inspection plan with trending of wear rate.

Case 12 - Superheater tube failure

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. Primary and platen superheater in the second pass; spray attemperation upstream of the PSH; SH metal temperature thermocouples on selected elements.

2. Incident summary - what happened

Flue gas temperature after the superheater began to fall and feed water consumption rose relative to steam flow. ID fan loading increased. Within a few hours a clear hissing was audible in the superheater region. The boiler was unloaded and tripped. Inspection revealed a burst superheater tube with a thick-lipped rupture, heavy internal oxide scale and evidence of localised starvation of that element caused by salt deposition. Two adjacent tubes had been eroded by the escaping steam jet.

2.1 Timeline and onset

- T-0:00 - Normal load; MS temperature 490 degC; SH metal temperatures near the upper end of their normal band on several elements.
- T+1:30 - Flue gas temperature after SH beginning to drop; ID fan current rising.
- T+3:00 - Feed water consumption higher than steam flow; drum level control working harder.
- T+4:30 - Hissing audible in the SH region during a field round.
- T+5:00 - Load reduction commenced.
- T+5:45 - Boiler tripped and isolated; leak located through manholes before depressurisation.
- Inspection - Thick-lipped burst on one SH element, heavy internal oxide scale, salt deposits restricting the element bore; steam-jet erosion on two adjacent tubes.

3. Evidence, in order of observation

- Falling flue gas temperature downstream of the superheater - steam entering the gas path absorbs heat and cools the gas.
- Rising ID fan loading - the leak adds mass flow to the gas path.
- High feed water consumption compared with steam flow.
- Hissing noise localised to the superheater region.
- SH tube metal temperature history showing sustained operation near or above the design limit on the affected element.
- History of drum level high excursions and of carry-over events, consistent with salt deposition inside superheater tubes.
- Excess air run high for long periods, giving elevated gas-side velocity and temperature.
- Physical evidence: thick-lipped rupture with wall thinning by creep, heavy internal oxide scale, deposit-restricted bore.

4. Components affected by the failure

- Superheater tubes - the failed element and the adjacent tubes damaged by the steam jet.
- Superheater headers and stub welds.
- Superheater supports and spacers.
- ID fan - overloaded by the added mass flow.
- APH and ESP - increased loading and moisture.

- Attemperator - unable to control temperature once flow distribution is disturbed.
- Turbine - risk of debris carry-over if the failure is not contained.

5. Initial hypotheses for the failure mechanism

- H1 - Long-term overheating (creep) from sustained high metal temperature due to inadequate steam flow through the element.
- H2 - Starvation of the individual element from internal salt / deposit blockage restricting steam flow.
- H3 - Fireside erosion by ash at high excess air.
- H4 - Short-term overheating from a single severe temperature excursion.
- H5 - Fireside corrosion from fuel-borne alkali and sulphur species.
- H6 - Water-wall slagging shifting heat to the superheater and raising gas temperature at the SH inlet.
- H7 - Original weld or material defect.

5.1 Failure modes observed

- Failure mode - creep rupture from long-term overheating (thick-lipped burst with heavy oxide scale).
- Failure mode - internal deposit accumulation causing flow starvation in one element.
- Consequential failure mode - steam-jet erosion of adjacent tubes.
- Consequential failure mode - loss of superheater duty and reduced final steam temperature after the leak.

6. Which evidence supports which hypothesis

- Heavy internal oxide scale is the definitive marker of long-term (rather than short-term) overheating and supports H1.
- The deposit-restricted bore in the failed element, with neighbouring elements clean, explains why that particular tube overheated and supports H2 as the cause of H1.
- The record of drum level high excursions and carry-over supplies the source of the salt deposits, linking H2 back to a known operational history.
- Uniform external wall condition with no directional grooving argues against H3.
- The absence of a single large excursion in the trend history and the presence of thick scale argue against H4.
- No molten-phase or laminated corrosion products were found on the fireside, weakening H5.
- Furnace exit gas temperature and water-wall absorption records were normal, weakening H6.

6.1 Most discriminating evidence

The most discriminating evidence is the internal condition of the failed element compared with its neighbours: heavy oxide scale and a deposit-restricted bore in the failed tube, with clean bores in the adjacent elements at the same gas temperature. This localises the cause to the steam side of that one element and rules out every gas-side explanation, which would have affected neighbouring tubes equally.

7. How competing hypotheses were ruled out

- H3 (fireside erosion) ruled out - the rupture was thick-lipped with wall thinning by creep deformation, not by external material loss; no directional erosion pattern was present and UT of adjacent tubes showed normal thickness.
- H4 (short-term overheating) ruled out - short-term overheat produces a thin-lipped, sharply-open fish-mouth with little oxide; here the scale was thick and the deformation was gradual.
- H5 (fireside corrosion) ruled out - no laminated or molten-phase corrosion products; ash analysis showed alkali within normal limits.
- H6 (water-wall slagging) ruled out - the neighbouring elements at the same gas temperature were undamaged, which is inconsistent with a bulk gas-temperature rise.

- H7 (material / weld defect) ruled out - the failure was in parent material remote from welds, and material certification and hardness checks were satisfactory.

8. Final failure mechanism determined

Creep rupture from long-term overheating of a single superheater element. Salt deposition inside that element, originating from earlier drum-level-high carry-over events, restricted the steam flow through it. With reduced internal cooling at unchanged gas-side heat input, the tube metal ran above its design temperature for an extended period; oxide scale grew, further raising metal temperature, and creep damage accumulated until the wall could no longer sustain the internal pressure.

9. Operational and maintenance context relevant to root cause

- Repeated drum level high and carry-over events had occurred without any subsequent steam-purity or superheater deposit assessment.
- Boiler water and spray water quality had been allowed to drift, and spray water was drawn from a source occasionally out of specification.
- SH tube metal temperature high alarms were annunciated but had no defined mandatory action.
- The boiler had been operated at high excess air to control bed temperature, raising gas temperature entering the superheater.
- No steam-side oxide-scale thickness measurement or creep-life assessment had ever been performed, even though the boiler had significant operating hours.
- The superheater had never been chemically cleaned on the steam side.

10. Final root cause

Physical root cause: internal salt deposition in one superheater element restricting steam flow and causing long-term overheating and creep rupture. Systemic root cause: carry-over events and poor spray-water and boiler-water quality were tolerated without any consequential inspection, and the maintenance strategy contained no oxide-scale or creep-life monitoring, so a slowly accumulating, entirely predictable damage mechanism ran to failure undetected.

11. Corrective actions to be taken

- Immediate: on noticing leakage, reduce load and trip the boiler; listen in the superheater region and try to locate the leakage through manholes before the boiler is depressurised.
- Immediate: after cooling, replace the failed element and inspect the adjacent tubes damaged by the steam jet, replacing any below minimum wall thickness.
- Short term: check for blocked and starved tubes across the bank; verify spray water quality and drum level control performance.
- Short term: control excess air to the design band to reduce gas temperature entering the superheater.
- Medium term: enforce boiler water and spray water chemistry limits, and correct the drum level control performance so that carry-over does not recur.
- Medium term: define mandatory operator action limits on SH metal temperature high alarms, including load reduction.
- Long term: introduce steam-side oxide-scale thickness measurement and creep-life assessment at overhaul for the highest-temperature elements, and chemically clean the superheater when scale exceeds the assessment threshold; consider selective element replacement based on the life assessment.

Case 13 - APH-PA and APH-SA tube failure

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. Tubular air pre-heaters on both PA and SA streams, located in the cold end of the gas path; APH by-pass dampers provided on the gas side.

2. Incident summary - what happened

Flue gas temperature after the air pre-heater began to fall while the oxygen percentage measured at the ESP inlet rose. PA and SA fan motor currents increased. Bed temperature rose slightly. The plant continued operating for several weeks with progressively worsening symptoms until a shutdown, when the APH-PA tube bundle was found with multiple perforated and thinned tubes at the cold end, showing external corrosion and internal erosion.

2.1 Timeline and onset

- T-0 weeks - Baseline: APH outlet gas temperature 150 degC, ESP inlet O₂ 4.5%, PA fan current normal.
- T+2 weeks - APH outlet gas temperature down to 138 degC; ESP inlet O₂ rising to 5.5%.
- T+4 weeks - PA and SA fan currents noticeably higher for the same delivered air flow; bed temperature drifting up.
- T+6 weeks - APH outlet gas temperature 128 degC; ESP inlet O₂ 6.5%; ID fan also loaded; coal consumption per unit of steam rising.
- T+7 weeks - Planned shutdown taken; APH-PA bundle inspected.
- Inspection - Multiple cold-end tubes perforated; external surfaces showing acid-dew-point corrosion; internal surfaces showing ash erosion at the tube inlets.

3. Evidence, in order of observation

- Falling flue gas temperature after the APH - air leaking into the gas stream dilutes and cools it.
- Rising oxygen percentage measured at the ESP inlet with no change in combustion air setting - the classic APH leakage signature.
- PA and SA fans overloaded: higher motor current for the same delivered air flow at the wind box.
- Rise in bed temperature, since less of the delivered primary air reaches the bed.
- ID fan loading increased by the extra mass flow through the back-pass.
- Rising specific coal consumption - loss of heat recovery.
- Cold-end metal temperature running below the acid dew point of the flue gas.
- Physical evidence: perforated and thinned cold-end tubes, external corrosion products, internal erosion at tube inlets.

4. Components affected by the failure

- APH-PA and APH-SA tube bundles, particularly the cold-end rows.
- APH casing, tube sheets and expansion joints.
- PA and SA fans and their motors - overloaded.
- ID fan - additional mass flow.
- ESP - altered gas volume, temperature and moisture, affecting collection efficiency.
- Bed and combustion system - reduced effective air, higher bed temperature.
- Ducting downstream - corrosion and ash deposition.

5. Initial hypotheses for the failure mechanism

- H1 - Cold-end corrosion from operating below the acid dew point (sulphur in fuel plus low metal temperature).
- H2 - Erosion of tubes by ash at the tube inlets.
- H3 - Mechanical failure - thermal fatigue at tube-to-tube-sheet joints, differential expansion.
- H4 - Faulty O₂ or gas temperature instrumentation giving a spurious leakage signature.
- H5 - Air leakage elsewhere in the duct or casing rather than through the APH tubes.
- H6 - Fouling / plugging of the APH rather than leakage.

5.1 Failure modes observed

- Failure mode - cold-end acid dew point corrosion causing tube wall loss and perforation.
- Failure mode - ash erosion at tube inlets thinning the tube ends.
- Consequential failure mode - air-to-gas leakage with fan overloading and loss of heat recovery.
- Consequential failure mode - bed temperature rise from reduced effective primary air.

6. Which evidence supports which hypothesis

- The simultaneous rise in ESP inlet O₂ and fall in APH outlet gas temperature is the defining combination for air-to-gas leakage across the APH, supporting H1, H2 and H3.
- PA and SA fan overloading for the same delivered wind box air flow confirms the leakage is on the air-to-gas path and not a duct-to-atmosphere leak, weakening H5.
- Cold-end metal temperature below the calculated acid dew point, together with the external corrosion products on the failed tubes, supports H1 as the dominant mechanism.
- Internal erosion confined to the tube inlets supports H2 as a secondary, location-specific contributor.
- Rising rather than falling gas-side differential pressure would have indicated fouling; here the differential pressure did not rise, weakening H6.
- The physically perforated tubes found at inspection refute H4 conclusively.

6.1 Most discriminating evidence

The most discriminating evidence is the location and appearance of the damage: perforation concentrated in the cold-end rows with external corrosion products, rather than uniform damage across the bundle. Erosion alone would concentrate damage at the gas-entry face and at tube inlets; mechanical fatigue would concentrate it at tube sheets. Cold-end corrosion is uniquely identified by damage at the coldest rows with acidic corrosion products.

7. How competing hypotheses were ruled out

- H3 (mechanical / thermal fatigue) ruled out - tube-to-tube-sheet joints were sound and the failures were mid-tube perforations in the cold rows, not joint cracking.
- H4 (instrumentation fault) ruled out - the physical perforations were found and the O₂ and temperature indications were independently verified by portable instruments before shutdown.
- H5 (duct / casing leakage) ruled out - fan overloading with the same wind box flow shows air is being lost from the air side into the gas side; a casing leak to atmosphere would not raise ESP inlet O₂.
- H6 (fouling) ruled out - gas-side differential pressure across the APH did not increase, and no significant deposit build-up was found.
- H2 (erosion) retained as a secondary mechanism - present at tube inlets, but insufficient by itself to explain perforations concentrated in the coldest rows.

8. Final failure mechanism determined

Cold-end acid dew point corrosion of the APH tubes. Sulphur in the coal produced sulphur trioxide, which combined with moisture in the flue gas to condense as sulphuric acid on tube surfaces running below the acid dew point in the cold-end rows. Progressive external wall loss, with ash erosion at the tube inlets as a

secondary contributor, perforated the tubes and allowed high-pressure primary and secondary air to leak directly into the flue gas stream.

9. Operational and maintenance context relevant to root cause

- Cold-end metal temperature was not being monitored or controlled; there was no steam coil air pre-heater or air by-pass strategy to keep the cold end above the acid dew point.
- Coal sulphur content had increased with a change of supplier, raising the acid dew point, and no acid dew point recalculation was made.
- The APH gas-side by-pass dampers were available but were not being used to control flue gas temperature.
- Low-load operation had become common, further depressing cold-end metal temperature.
- APH leakage was never quantified; there was no periodic leakage test and no trending of the O₂ rise across the APH.
- The symptoms were visible for weeks and were normalised as a gradual drift rather than treated as a developing failure.

10. Final root cause

Physical root cause: sulphuric acid condensation on cold-end APH tube surfaces operating below the acid dew point, causing progressive corrosion and perforation. Systemic root cause: the cold end was not protected or monitored - no cold-end metal temperature control, no recalculation of the acid dew point after a fuel sulphur increase, and no periodic APH leakage measurement - so a known, temperature-controlled corrosion mechanism was allowed to run for the life of the bundle.

11. Corrective actions to be taken

- Immediate: control flue gas temperature by using the PA-APH gas-side by-pass to raise cold-end metal temperature above the acid dew point.
- Immediate: reduce coal feeding as required to maintain the O₂ percentage, and control bed temperature, since less primary air is now reaching the bed.
- Immediate: if the leakage of tubes is severe, stop the boiler and plug or replace the affected tubes.
- Short term: quantify APH leakage by an O₂ rise measurement across the APH, and repeat it periodically as a trended parameter.
- Medium term: recalculate the acid dew point for the actual fuel sulphur and set a minimum cold-end metal temperature limit; use APH gas by-pass and, if justified, a steam coil air pre-heater to hold it.
- Medium term: control coal sulphur through supplier specification or blending.
- Long term: replace the cold-end rows with corrosion-resistant material (for example, Corten or enamelled tubes) as a design change, and add cold-end metal temperature measurement with an alarm on approach to the dew point.

Case 14 - Loss of coal feeder (DCF) from a common electrical cause

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. Each DCF is driven by a VFD, all six VFDs fed from a common MCC bus section; DCF trips are also initiated by bed temperature high (940 degC), bed temperature low (320 degC), furnace pressure high or low, drum level high or low, and PA fan trip.

2. Incident summary - what happened

A single-phasing fault developed on the incomer of the MCC bus feeding all six coal feeder VFDs. All DCFs stopped within seconds. Coal feed to the bed ceased entirely. Boiler pressure fell rapidly, bed temperature dropped, superheater steam temperature fell sharply and drum level and furnace pressure fluctuated. The turbine load had to be shed and the boiler was taken to a controlled shutdown before the bed slumped uncontrollably.

2.1 Timeline and onset

- T-0:00 - 90% MCR, all six DCFs running, all bed temperatures 860-880 degC.
- T+0:00:10 - Single-phasing on the MCC incomer; all six VFDs trip on phase loss / under-voltage.
- T+0:00:30 - Coal feed to all compartments zero; PA and SA fans still running at full flow.
- T+0:02 - Bed temperatures falling rapidly across all compartments; steam pressure falling; SH steam temperature falling sharply.
- T+0:04 - Drum level fluctuating on the pressure transient; furnace pressure fluctuating.
- T+0:06 - Turbine load reduced; PA flow cut back to reduce bed cooling; attempt to restart feeders unsuccessful (MCC fault).
- T+0:10 - Controlled shutdown; bed slumped in a controlled manner; oil burner support taken to preserve bed temperature where possible.
- T+1:00 - Electrical fault located: loose / burnt connection on one phase of the MCC incomer.

3. Evidence, in order of observation

- Simultaneous loss of all six feeders within seconds - a common-cause signature that no process cause could produce.
- VFD fault codes recording phase loss / under-voltage rather than mechanical overload.
- MCC incomer protection operation recorded in the electrical event log.
- No preceding bed temperature, drum level or furnace pressure excursion that could have tripped the feeders on process interlock.
- PA fan running normally throughout, excluding the PA-fan-trip interlock as the cause.
- Rapid fall in boiler pressure and in bed temperature across all compartments equally.
- Sharp fall in superheater steam temperature and variation in drum level and furnace pressure.
- Physical evidence: burnt and loose connection on one phase of the MCC incomer.

4. Components affected by the failure

- All six drag chain feeders and their VFD drives.
- MCC bus section and incomer - the failed component.
- Bed material and distributor plate - risk of defluidisation and of unburnt fuel accumulation.
- In-bed evaporator coils - thermal shock from a rapid bed temperature fall.

- Superheater - sharp temperature drop and thermal cycling.
- Turbine - forced load shedding and steam temperature transient.
- Drum and drum internals - pressure and level transient.

5. Initial hypotheses for the failure mechanism

- H1 - Common electrical supply failure to all feeder drives (MCC / incomer / bus fault).
- H2 - Process interlock trip - bed temperature high or low, furnace pressure high or low, drum level high or low, syphon temperature, or main steam temperature high.
- H3 - PA fan trip causing the feeder trip interlock to act.
- H4 - VFD fault common to the drives (for example, a shared control supply or a common firmware fault).
- H5 - Mechanical jamming of all feeders - implausible but formally considered.
- H6 - Operator error - all feeders stopped from DCS.

5.1 Failure modes observed

- Common-cause failure mode - loss of electrical supply to all coal feeders simultaneously.
- Component failure mode - loose / degraded electrical connection leading to single-phasing.
- Consequential failure mode - total loss of heat input, bed temperature collapse and boiler pressure loss.
- Consequential failure mode - thermal transient on superheater and turbine.

6. Which evidence supports which hypothesis

- Simultaneity across six independent mechanical drives is by itself near-conclusive for a common-cause supply failure, supporting H1 and H4 and excluding H5.
- VFD fault codes reading phase loss / under-voltage discriminate H1 from H4 - a drive-internal or firmware fault would not report an incoming supply abnormality.
- The absence of any process excursion before the trip excludes H2 - process interlocks leave a clear trace on the process trend, and none was present.
- PA fan status normal throughout excludes H3.
- DCS operator command logs show no stop command, excluding H6.
- The physically found burnt connection on the incomer confirms H1 directly.

6.1 Most discriminating evidence

The most discriminating evidence is the VFD fault code content. All six drives reporting an incoming supply abnormality (phase loss / under-voltage) at the same instant distinguishes a supply-side common-cause failure from a process interlock, from an operator action and from a drive-internal fault, each of which would leave a completely different trace.

7. How competing hypotheses were ruled out

- H2 (process interlock) ruled out - bed temperatures, drum level, furnace pressure and steam temperature were all inside their normal bands at the moment of the trip; the excursions followed the trip, not preceded it.
- H3 (PA fan trip) ruled out - both PA fans continued running with normal current and PA header pressure was maintained.
- H4 (common VFD fault) ruled out - the drives reported a supply abnormality, not an internal fault, and they restarted normally once the supply was restored.
- H5 (mechanical jamming) ruled out - no shear pins failed and all chains rotated freely when tested; simultaneous mechanical jamming of six independent feeders is not credible.
- H6 (operator error) ruled out - DCS command logs show no stop command from any operator station.

8. Final failure mechanism determined

Total loss of fuel input due to a common-cause electrical failure: a degraded connection on one phase of the MCC incomer produced single-phasing, causing all six coal feeder VFDs to trip on phase loss. Because all feeders shared one bus section without segregation, a single connection defect removed the entire fuel feed, and the thermal inertia of the bed then decayed with no replacement heat input.

9. Operational and maintenance context relevant to root cause

- All six feeder drives were fed from a single MCC bus section with no segregation between odd and even compartments - a single-point-of-failure design.
- Thermographic (infrared) survey of MCC connections was not part of the electrical PM schedule.
- Torque-checking of bus and incomer connections had not been carried out since commissioning.
- There was no automatic action to reduce PA flow or to take burner support on total loss of coal feed, so the bed was actively cooled while it lost its heat source.
- The operating procedure for loss of coal feeder addressed the loss of one feeder, not the loss of all feeders.
- Standby / emergency fuel (oil burner) availability had not been proof-tested recently.

10. Final root cause

Physical root cause: single-phasing of the MCC incomer from a degraded electrical connection, tripping all coal feeder VFDs simultaneously. Systemic root causes: (a) a design in which all six feeders depend on one un-segregated bus section, so a single connection defect is a total-loss-of-fuel event, and (b) an electrical maintenance programme with no thermography and no connection-torque verification, so a slowly degrading joint could not be detected before failure.

11. Corrective actions to be taken

- Immediate: control boiler pressure by reducing load on the turbine; control superheater steam temperature by closing the spray valve; control furnace draught and drum level.
- Immediate: check the electrical fault and the emergency stop push buttons, check the VFD fault, and restart the coal feeders only after establishing the cause of failure.
- Immediate: cut back PA flow so the bed is not actively cooled, and take oil burner support to preserve bed temperature if restart is likely within the ignition window.
- Short term: repair the MCC incomer connection and inspect all connections on the bus section.
- Medium term: introduce annual infrared thermography and connection-torque verification on all MCC and switchgear connections in the boiler auxiliaries.
- Medium term: extend the operating procedure to cover total loss of coal feed, including PA flow reduction, burner support and controlled bed slumping.
- Long term: segregate feeder drives across two independent bus sections so that no single electrical fault can remove all fuel feed, and review the same single-point-of-failure exposure for other critical auxiliaries.

Case 15 - Hot bed material leakage from PA lines and back fire in PA lines

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. Primary air is distributed to each compartment wind box through PA lines with individual air dampers; bed ash drains are provided per compartment. Back fire occurs when hot bed material or flame propagates backwards into the air line during low or interrupted PA flow.

2. Incident summary - what happened

A PA line air damper to compartment 3 was inadvertently left throttled after a maintenance activity. With reduced air velocity in that line, hot bed material back-flowed into the vertical portion of the PA line, and glowing material was observed at the line. Bed temperature in compartment 3 became erratic. The leakage of hot material increased and the compartment was cut out; coal feeding to it was stopped and its wind box damper closed until the line could be cleared.

2.1 Timeline and onset

- T-0:00 - Maintenance completed on the compartment 3 PA line; damper left partly closed and not verified on restoration.
- T+0:30 - Compartment 3 bed temperature becoming erratic; PA flow to that compartment low.
- T+0:50 - Glowing / hot material observed at the vertical portion of the PA line; local temperature at the line rising.
- T+1:05 - Bed temperature in compartment 3 rising in places and falling in others - localised defluidisation.
- T+1:15 - Coal feeding to compartment 3 cut, bed temperature allowed to fall, wind box damper closed.
- T+1:25 - Air damper of that line confirmed and opened; line cleared of accumulated material from the vertical portion.
- T+2:00 - Compartment restored progressively; boiler continued at reduced load throughout.

3. Evidence, in order of observation

- PA flow to compartment 3 lower than to the other compartments, with the same header pressure - localising the restriction to that line.
- Erratic and non-uniform bed temperature in compartment 3, indicating loss of uniform fluidisation.
- Glowing hot material visible at the vertical portion of the PA line.
- Rising surface temperature of the PA line and of the wind box in that region.
- PA header pressure normal - the PA fan and the header are healthy, so the problem is downstream in the branch.
- Maintenance record showing recent work on that line with no restoration check on the damper position.
- No abnormality in the other five compartments - a single-branch problem.

4. Components affected by the failure

- PA line to the affected compartment, particularly the vertical portion where material accumulates.
- PA line air damper and its position indication.
- Wind box and distributor nozzles of that compartment - blockage and erosion.
- Bed material in that compartment - localised defluidisation and possible sintering.

- In-bed coils in the affected compartment - non-uniform heat transfer and thermal stress.
- Bed ash drain line of that compartment.
- Personnel safety - hot material discharge from the line is a burn hazard.

5. Initial hypotheses for the failure mechanism

- H1 - Low or interrupted PA flow in one branch allowing hot bed material to back-flow into the line.
- H2 - PA fan trip or header pressure loss.
- H3 - Distributor nozzle failure or missing nozzle caps allowing bed material into the wind box.
- H4 - Excessive bed height increasing the static head against the air supply.
- H5 - Bed ash drain valve leaking or left open, drawing material into the line.
- H6 - Coal feeding to a poorly fluidised compartment causing local sintering and defluidisation.

5.1 Failure modes observed

- Failure mode - back-flow of hot bed material into the primary air line (back fire).
- Human / procedural failure mode - damper left throttled after maintenance without restoration verification.
- Consequential failure mode - localised defluidisation and non-uniform bed temperature.
- Hazard mode - hot material discharge at the line and burn risk to personnel.

6. Which evidence supports which hypothesis

- Low PA flow confined to one branch with normal header pressure is direct support for H1 and excludes H2.
- The maintenance record showing recent work on that line and no restoration check supplies the mechanism for the throttled damper, supporting H1.
- Erratic bed temperature in only that compartment supports the localised loss of fluidisation and is consistent with H1, H3 and H4.
- Bed differential pressure being normal across all compartments weakens H4.
- The bed ash drain valve was verified closed and cold, weakening H5.
- Coal feeding to compartment 3 had been normal until the temperature became erratic, so H6 is a consequence rather than a cause.

6.1 Most discriminating evidence

The most discriminating evidence is low PA flow in a single branch while the PA header pressure remains normal. This isolates the restriction to that branch and, combined with the damper position found after the event, identifies the throttled damper as the cause; a fan, header, bed-height or drain problem would affect more than one compartment or would show in header pressure.

7. How competing hypotheses were ruled out

- H2 (PA fan / header loss) ruled out - PA header pressure and both PA fan currents were normal, and the other five compartments were unaffected.
- H3 (nozzle failure) ruled out for this event - the wind box was inspected at the next opportunity and the nozzle caps were intact; the material had entered through the line, not through failed nozzles.
- H4 (excessive bed height) ruled out - bed differential pressure was within the normal band on all compartments including the affected one.
- H5 (bed ash drain) ruled out - the drain valve was verified closed and its line was cold.
- H6 (local sintering) ruled out as the initiator - it developed after the fluidisation was already lost and is therefore a consequence.

8. Final failure mechanism determined

Back-flow of hot bed material into the primary air line because the air velocity in that branch fell below the value needed to hold the bed pressure. With the damper throttled, the dynamic pressure in the line could no longer oppose the static head of the fluidised bed, so hot particles migrated backwards into the line and accumulated in its vertical portion, locally defluidising the compartment and creating a back-fire condition.

9. Operational and maintenance context relevant to root cause

- There was no post-maintenance restoration checklist requiring verification of damper positions before returning a line to service.
- PA line dampers have local manual operation with no position feedback in DCS, so a throttled damper is invisible to the control room.
- Compartment-wise PA flow was not measured or trended; only the header pressure was monitored.
- The operating procedure for hot bed material leakage existed but was not routinely rehearsed.
- No temperature monitoring existed on the PA lines, so back-flow could only be detected by a field observation.
- Bed material charging practice and bed height control were informal.

10. Final root cause

Physical root cause: insufficient primary air velocity in one branch line, allowing hot bed material to back-flow into it. Systemic root cause: absence of a post-maintenance restoration verification for damper positions, combined with no compartment-wise PA flow measurement and no PA line temperature monitoring, so a throttled damper could neither be prevented nor detected before it produced a back-fire.

11. Corrective actions to be taken

- Immediate: observe bed temperatures constantly and charge bed material if required.
- Immediate: cut down coal feeding to that compartment, let the bed temperature drop and close the wind box damper.
- Immediate: check that the air damper of that line is open, and clear the line if the leakage is from the vertical portion.
- Immediate: if the leakage increases further, stop the boiler; do not persist with a compartment that is discharging hot material.
- Immediate: barricade the area and keep personnel clear of the leaking line - hot bed material is a serious burn hazard.
- Short term: introduce a post-maintenance restoration checklist covering every damper, valve and isolation returned to service, signed by both the maintenance and the operations shift in charge.
- Medium term: provide compartment-wise PA flow indication and surface temperature monitoring on the PA lines, with alarms.
- Long term: provide position feedback for the PA line dampers in DCS, and review the line geometry so that the vertical portion is less prone to material accumulation.

Case 16 - Boiler Feed Pump failure

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. Two motor-driven multi-stage BFPs (1 working, 1 standby) taking suction from the deaerator; auto-start of standby on discharge header pressure low and on running pump trip; minimum recirculation valve on each pump; deaerator level alarm low at 20%.

2. Incident summary - what happened

The deaerator level fell below the low alarm because of a condensate transfer pump problem. The running BFP began to cavitate, its vibration rose beyond 9.3 mm/s, pump bearing temperature climbed above 110 degC and the mechanical seal started leaking. The pump tripped on vibration. The standby pump started on auto but, with the deaerator level still low, it also began to cavitate. Drum level fell and the boiler had to be unloaded rapidly. Inspection showed cavitation damage on the first-stage impeller and a damaged mechanical seal.

2.1 Timeline and onset

- T-0:00 - Normal operation; deaerator level 55%; BFP-A running; suction pressure normal.
- T+0:20 - Condensate transfer pump problem; deaerator level falling.
- T+0:35 - Deaerator level low alarm at 20%; BFP-A suction pressure falling; NPSH margin eroding.
- T+0:42 - BFP-A vibration rising above 5 mm/s and noise audible; discharge flow becoming unsteady.
- T+0:50 - Vibration above 9.3 mm/s, pump bearing temperature above 110 degC, mechanical seal leaking; BFP-A trips on vibration.
- T+0:51 - BFP-B starts on auto; drum level dipping; BFP-B also cavitating with the deaerator level still low.
- T+0:58 - Load reduced rapidly to reduce feed demand; deaerator level restored by cross-connecting a standby condensate pump.
- T+1:30 - Feed system stabilised; BFP-A isolated for inspection.
- Inspection - Cavitation erosion on the first-stage impeller eye; mechanical seal faces damaged; suction strainer partially choked.

3. Evidence, in order of observation

- Deaerator level falling below the low alarm ahead of every pump symptom - the earliest indication in the sequence.
- Falling BFP suction pressure and reduced NPSH available.
- Audible cavitation noise and crackling at the pump.
- Rising vibration, from normal to above the 9.3 mm/s trip level.
- Rising pump bearing temperature above 110 degC and motor winding temperature approaching its limit.
- Mechanical seal leakage.
- Unsteady discharge flow and falling discharge header pressure.
- Differential pressure across the suction strainer higher than normal.
- Physical evidence: cavitation erosion at the first-stage impeller eye; damaged seal faces; partially choked strainer.

4. Components affected by the failure

- BFP first-stage impeller and casing - cavitation erosion.
- BFP mechanical seal and seal faces.
- BFP bearings - loading from unstable flow and vibration.
- BFP motor - winding and bearing temperature excursion.
- Suction strainer.
- Minimum recirculation valve and its control.
- Deaerator and the condensate transfer system - the upstream origin.
- Boiler drum and water walls - at risk from the consequent low drum level.

5. Initial hypotheses for the failure mechanism

- H1 - Loss of NPSH from a low deaerator level, causing cavitation.
- H2 - Choked suction strainer restricting flow.
- H3 - Mechanical failure of bearings or of the rotor - misalignment or unbalance.
- H4 - Minimum recirculation valve failing to open, causing low-flow operation and overheating.
- H5 - Electrical fault - motor protection relay operation, low motor bearing oil pressure.
- H6 - Discharge-side restriction increasing pump load.
- H7 - Deaerator water temperature too high for the available static head (flashing at suction).

5.1 Failure modes observed

- Failure mode - cavitation erosion of the impeller from inadequate NPSH.
- Failure mode - mechanical seal damage secondary to unstable operation and vibration.
- Consequential failure mode - standby pump exposed to the same condition (common-cause failure of both trains).
- Consequential functional failure - loss of feed water supply and falling drum level.

6. Which evidence supports which hypothesis

- The deaerator level fall preceding the pump symptoms is the causal ordering that supports H1 above all others.
- Cavitation noise, unsteady discharge and impeller-eye erosion are the classic NPSH-deficiency triad, again supporting H1.
- The standby pump exhibiting the same behaviour immediately on start is strong evidence for a system-side (suction) cause rather than a pump-side cause, weakening H3, H4, H5 and H6 all at once.
- The partially choked strainer supports H2 as an aggravating contributor by further reducing suction pressure.
- Deaerator temperature and pressure were at saturation as designed and had not changed, giving no independent support to H7.
- Bearing temperature and seal leakage rose after the vibration, marking them as consequences rather than causes.

6.1 Most discriminating evidence

The most discriminating evidence is that the standby pump replicated the symptoms within seconds of starting. A defect internal to one pump cannot transfer to another; identical behaviour in two independent machines points unambiguously to the shared suction condition, and therefore to loss of NPSH from the low deaerator level.

7. How competing hypotheses were ruled out

- H3 (mechanical / alignment) ruled out - alignment and balance records were current, vibration had been normal until the deaerator level fell, and the standby machine behaved identically.
- H4 (recirculation valve) ruled out - the minimum recirculation valve was confirmed open with flow, and the pump was not in a low-flow condition; it was in an unstable-suction condition.

- H5 (electrical) ruled out - the trip was initiated by the vibration protection, not by the motor protection relay; motor current was normal until the flow became unstable.
- H6 (discharge restriction) ruled out - discharge header pressure was falling, not rising, which is the opposite of a discharge restriction.
- H7 (deaerator temperature) ruled out - deaerator pressure and temperature were at their normal saturation relationship; the NPSH loss came from the level, not from the temperature.
- H2 (choked strainer) retained as a contributor - it reduced the already-marginal suction pressure but was not sufficient alone, since the pump had been running satisfactorily with that strainer condition at normal deaerator level.

8. Final failure mechanism determined

Cavitation of the boiler feed pump from loss of net positive suction head. A condensate transfer failure allowed the deaerator level to fall, reducing the static head available at the pump suction below the NPSH required; the partially choked suction strainer removed the remaining margin. Vapour formed and collapsed at the first-stage impeller eye, eroding the impeller, destabilising the rotor, damaging the mechanical seal and finally tripping the pump on vibration. The standby pump, sharing the same suction source, cavitated on start for the same reason.

9. Operational and maintenance context relevant to root cause

- Deaerator level low alarm existed but there was no interlock reducing boiler load or tripping the feed pump before cavitation damage occurred.
- The condensate transfer system had a single working pump with an unreliable standby; its failure was a known but unaddressed risk.
- Suction strainer differential pressure was not monitored and strainers were cleaned only during shutdown.
- Both BFPs draw from a single deaerator with no independent suction source, so any suction-side disturbance is a common-cause event.
- Vibration and bearing temperature trends were available but were not being reviewed as condition-monitoring data; only the trip values were used.
- The auto-start of the standby pump was correctly functional, but starting a second pump into a degraded suction condition made the situation worse rather than better.

10. Final root cause

Physical root cause: loss of NPSH at the boiler feed pump suction caused by a low deaerator level, aggravated by a partially choked suction strainer, leading to cavitation damage. Systemic root cause: the deaerator level was a single-point common-cause vulnerability for both feed pump trains, protected only by an alarm and not by an interlock or by a reliable condensate transfer system, and the suction strainer condition was neither monitored nor maintained on condition.

11. Corrective actions to be taken

- Immediate: start the standby pump if it does not start on auto, and adjust the load to maintain drum level within the available feed water flow.
- Immediate: check the deaerator level and, if it falls below the safe limit, stop the feed pump immediately rather than allowing it to cavitate.
- Immediate: check the BFP suction strainer and clean it if found choked; check the proper operation of the recirculation valve and the cooling water valve.
- Short term: analyse and rectify the fault in the main feed pump, restore the impeller and mechanical seal, and put the pump back on auto so the standby function is genuinely available.
- Short term: restore the condensate transfer standby pump to service.
- Medium term: interlock deaerator level low to an automatic boiler load runback and to a feed pump trip before the cavitation threshold, so that the alarm is not the only defence.

- Medium term: monitor and trend suction strainer differential pressure, BFP vibration and bearing temperature as condition-monitoring parameters rather than as trip inputs only.
- Long term: review the single-deaerator common-cause exposure for the feed system and provide adequate deaerator storage volume or an alternative suction source.

Case 17 - Loss of ID fan and furnace pressurisation

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. 2x50% ID fans with inlet damper / vane control; furnace pressure trip at +100 mmwc; boiler trips on furnace pressure high; ID fan trips on motor protection, bearing temperature high and drive (MCC) fault.

2. Incident summary - what happened

The running ID fan-A tripped on high bearing temperature after its cooling water supply was inadvertently isolated during work on an adjacent line. ID fan-B was running at part load and could not absorb the full gas flow. Furnace pressure rose rapidly from -10 mmwc into the positive range; hot gas puffed from inspection doors and expansion joints. The boiler tripped on furnace pressure high. Inspection found the ID-A fan bearing damaged and the fan casing distorted at the inspection door seal.

2.1 Timeline and onset

- T-0:00 - Both ID fans in service; furnace pressure -10 mmwc; boiler at 88% MCR.
- T-0:40 - Cooling water line work on an adjacent system; ID-A fan bearing cooling water isolated inadvertently.
- T+0:00 - ID-A bearing temperature rising; high alarm annunciated.
- T+0:06 - ID-A trips on bearing temperature high.
- T+0:06:10 - Furnace pressure rising rapidly; ID-B inlet vanes open to maximum but capacity is insufficient.
- T+0:06:40 - Furnace pressure exceeds +100 mmwc; hot gas puffing observed at inspection doors; boiler trips on furnace pressure high.
- T+0:08 - FD and PA fans tripped by interlock; coal feeders tripped; bed slumped.
- Inspection - ID-A bearing damaged by overheating; casing distortion at an inspection door seal; cooling water valve found closed and not tagged.

3. Evidence, in order of observation

- ID-A bearing temperature rising steadily before the trip, with no rise in vibration or motor current - a cooling problem rather than a mechanical or load problem.
- Cooling water flow to the ID-A bearing absent, with the isolating valve found closed and untagged.
- ID-A trip initiated by bearing temperature high, recorded in the event log.
- Furnace pressure rising immediately after the ID-A trip - a clear causal ordering.
- ID-B inlet vanes at maximum with insufficient capacity, and ID-B motor current at its limit.
- Hot gas puffing from inspection doors and expansion joints - physical confirmation of furnace pressurisation.
- Boiler trip on furnace pressure high recorded.
- Physical evidence: damaged ID-A bearing showing thermal degradation of the lubricant.

4. Components affected by the failure

- ID fan-A bearing, shaft and impeller - the failed component.
- ID fan casing and inspection door seals - distortion from the pressure excursion.
- Furnace walls, ducting, expansion joints and enclosures - pressurisation loading.
- APH and ESP - exposed to the pressure transient.

- FD, PA and SA fans - tripped by interlock.
- Bed and coal feeders - forced slumping.
- Personnel in the boiler area - hot gas puffing is a direct burn hazard.

5. Initial hypotheses for the failure mechanism

- H1 - ID fan bearing failure from loss of cooling water.
- H2 - Electrical motor protection operation - overload, earth fault, MCC fault.
- H3 - Fan overloading from a gas-path restriction or from excessive gas mass flow (for example, APH leakage).
- H4 - Mechanical failure of the fan - unbalance, blade erosion, coupling failure.
- H5 - Control failure - inlet vane control fault causing loss of draught control.
- H6 - Faulty bearing temperature indication causing a spurious trip.
- H7 - Furnace pressure transmitter fault causing a spurious boiler trip.

5.1 Failure modes observed

- Component failure mode - ID fan bearing thermal degradation and seizure from loss of cooling.
- Human / procedural failure mode - isolation of a cooling water line serving running equipment without permit control or tagging.
- Consequential failure mode - furnace pressurisation with hot gas escape.
- Consequential failure mode - fan casing and door seal distortion.

6. Which evidence supports which hypothesis

- Bearing temperature rising with vibration and motor current unchanged is the signature of a cooling loss rather than of a mechanical or load fault, supporting H1 and weakening H3 and H4.
- The physically found closed and untagged cooling water valve, together with the timing of the adjacent work, supplies the direct causal mechanism for H1.
- Furnace pressure rising only after the ID-A trip establishes the ordering and rules out any hypothesis in which the pressure excursion came first.
- ID-B at maximum vane opening with its motor at current limit explains why the second fan could not compensate - a capacity issue, not a control fault, which weakens H5.
- Physical hot gas puffing refutes H7 - the pressurisation was real.
- Multiple bearing temperature elements in agreement, and the physically damaged bearing, refute H6.

6.1 Most discriminating evidence

The most discriminating evidence is the rise in bearing temperature with no accompanying rise in vibration or motor current. Mechanical degradation (unbalance, misalignment, bearing wear) raises vibration first; overloading raises motor current first. A rise in temperature alone points specifically to loss of heat removal, i.e. loss of cooling water, and the closed valve confirmed it.

7. How competing hypotheses were ruled out

- H2 (electrical) ruled out - the trip was initiated by the bearing temperature protection; motor protection relays had not operated and motor current was normal until the trip.
- H3 (gas-path restriction / overload) ruled out - ID-A motor current and gas-side differential pressures were normal before the bearing temperature rise.
- H4 (mechanical failure) ruled out as the initiator - vibration was normal until the temperature excursion; the bearing damage found afterwards was thermal, consistent with lubricant degradation rather than with wear or unbalance.
- H5 (control fault) ruled out - both fans responded correctly to their vane demands; ID-B simply lacked the capacity to take the whole gas flow at that load.

- H6 (temperature indication fault) ruled out - independent elements agreed and the bearing was physically damaged.
- H7 (pressure transmitter fault) ruled out - hot gas physically escaped from the enclosure, confirming a real pressurisation.

8. Final failure mechanism determined

Thermal failure of the ID fan bearing from loss of bearing cooling water, leading to a fan trip. With one ID fan out of service and the remaining fan already at its capacity limit for the prevailing load, the boiler lost its ability to maintain negative furnace draught. Gas continued to be produced at the same rate with insufficient extraction, so furnace pressure rose rapidly into the positive range until the furnace pressure high protection tripped the boiler; in the interval, hot gas escaped through the weakest seals.

9. Operational and maintenance context relevant to root cause

- The cooling water isolation was performed without a permit-to-work covering the running ID fan and without valve tagging - a work-control failure.
- There was no flow switch or low-flow alarm on the ID fan bearing cooling water line, so the loss of cooling was invisible until the temperature rose.
- The unit was being operated at a load at which a single ID fan could not maintain draught, with no automatic load runback on loss of one ID fan.
- Bearing temperature trends were not reviewed as condition data; only the alarm and trip values were used.
- Boiler enclosure inspection doors and expansion joints had not been inspected for integrity within the maintenance cycle.
- Operators had no defined action for a single-ID-fan condition other than to watch furnace pressure.

10. Final root cause

Physical root cause: ID fan bearing failure from loss of cooling water flow. Systemic root causes: (a) a work-control system that permitted isolation of a service supporting running critical equipment without permit or tagging, (b) absence of a low-flow alarm on the bearing cooling water, so the loss was undetectable until damage occurred, and (c) absence of an automatic load runback on loss of one ID fan, so the plant was operating with no capability to survive a single-fan failure.

11. Corrective actions to be taken

- Immediate: check for an emergency or field stop, rectify the electrical fault if any, and check the fan and motor bearing temperatures before restarting.
- Immediate: restart the ID fan only after checking the cause of tripping and taking corrective action; do not restart into an unresolved fault.
- Immediate: restore and prove the bearing cooling water supply, and verify no other cooling line has been isolated.
- Short term: repair the damaged bearing and inspect the shaft journal; inspect and repair the distorted casing and door seals before returning the fan to service.
- Short term: brief operators that on loss of one ID fan, load must be reduced immediately to what the remaining fan can handle, and that the boiler should be allowed to trip on furnace draught high rather than force-run pressurised.
- Medium term: install flow switches and low-flow alarms on all fan bearing cooling water lines.
- Medium term: implement an automatic load runback interlocked to loss of one ID fan, and review the single-fan capacity margin at each load band.
- Long term: strengthen permit-to-work and lock-out / tag-out discipline for any isolation affecting running equipment, and include enclosure and expansion joint integrity in the periodic inspection plan.

Case 18 - Loss of PA fan with bed defluidisation and back-fire risk

1. Asset context

Asset model and make: Boiler-1, 67 TPH / 67 kg/cm²(g) / 495 degC AFBC bi-drum boiler (Indian OEM class), 6-compartment bed with 6 DCFs, 2x50% ID / FD(SA) / PA fans, tubular APH-PA and APH-SA, feeding a 15 MW captive TG set.

Operating context: Base-load captive operation at 85-100% MCR, 3-shift running, blended Indian and imported coal, design bed temperature 850-900 degC, drum level normal 50%, furnace draught -5 to -20 mmwc, main steam temperature alarm 550 degC. 2x50% PA fans; PA fan trips on motor protection, PA outlet pressure high (above 1800 mmwc), motor winding temperature high (above 140 degC), motor bearing temperature high (above 150 degC), drum level high or low, furnace pressure high, MCC fault, run feed-back off, and on both ID or both SA fans tripping. Coal feeders trip on PA fan trip.

2. Incident summary - what happened

PA fan-A tripped on motor winding temperature high after a cooling air passage was found choked with coal dust. PA fan-B was on standby but its auto-start did not occur because the run feed-back contact was defective. Primary air to the bed was lost. All coal feeders tripped on the PA fan trip interlock as designed. The bed defluidised and hot material began to migrate into the PA lines. Furnace draught went low and the boiler tripped. Restoration required clearing the PA lines and re-charging bed material.

2.1 Timeline and onset

- T-0:00 - PA fan-A running, PA fan-B on standby; PA header pressure normal; six compartments in service.
- T-0:30 - PA fan-A motor winding temperature rising; alarm at 140 degC.
- T+0:00 - PA fan-A trips on motor winding temperature high.
- T+0:00:05 - PA fan-B auto-start command issued; fan does not start - run feed-back contact defective, start sequence aborts.
- T+0:00:10 - PA header pressure collapses; primary air to all compartments lost.
- T+0:00:15 - All coal feeders trip on the PA fan trip interlock (as designed).
- T+0:01 - Bed defluidises; bed temperature indication becomes erratic; hot bed material starts entering the PA lines.
- T+0:02 - Furnace draught low; boiler trips.
- T+0:30 - PA fan-B started manually; PA lines cleared of migrated material; bed material re-charged before restart.
- Inspection - PA fan-A motor cooling air passages choked with coal dust; PA fan-B run feed-back contact found defective.

3. Evidence, in order of observation

- PA fan-A motor winding temperature rising over 30 minutes before the trip - a developing thermal fault, not a sudden electrical fault.
- PA fan-A motor current normal, ruling out overload as the cause of heating.
- Trip initiated by motor winding temperature high, recorded in the event log.
- PA fan-B auto-start command present in DCS with no fan start and no run feed-back - a start-permissive failure.
- Collapse of PA header pressure and loss of primary air flow to all compartments.
- All coal feeders tripping simultaneously on the PA fan trip interlock - the designed protective response.
- Erratic bed temperature indication and low furnace draught following the loss of fluidising air.
- Physical evidence: choked motor cooling air passages full of coal dust; defective run feed-back contact on PA fan-B.

4. Components affected by the failure

- PA fan-A motor - winding thermal damage and insulation life consumption.
- PA fan-B start circuit - run feed-back contact.
- Bed material and distributor plate - defluidisation, settling and possible sintering.
- PA lines and wind box - hot material migration and back-fire risk.
- Distributor nozzles - blockage by settled material.
- Coal feeders - tripped by interlock.
- Boiler as a whole - forced trip and a difficult restart requiring bed re-charging.

5. Initial hypotheses for the failure mechanism

- H1 - PA fan motor thermal failure from blocked cooling air passages.
- H2 - Motor electrical fault - winding insulation failure, earth fault, motor protection relay operation.
- H3 - Fan overload from a high PA outlet pressure condition (above 1800 mmwc) caused by choked PA lines or closed dampers.
- H4 - Motor bearing failure causing heating.
- H5 - MCC fault or loss of supply.
- H6 - Consequential trip from another interlock - drum level, furnace pressure, ID fan trip or SA fan trip.
- H7 - Failure of the standby PA fan auto-start (a separate, second failure).

5.1 Failure modes observed

- Component failure mode - motor winding overheating from loss of cooling air path (fouling by coal dust).
- Hidden failure mode - standby PA fan auto-start unavailable due to a defective run feed-back contact.
- Consequential failure mode - bed defluidisation and hot material migration into the PA lines.
- Consequential failure mode - total loss of fuel input via the designed coal feeder trip interlock.

6. Which evidence supports which hypothesis

- Winding temperature rising while motor current stays normal is the signature of a cooling deficiency and supports H1 while weakening H2 and H3.
- PA outlet pressure remaining below the 1800 mmwc trip value throughout, and PA line dampers being in their normal positions, weakens H3.
- Motor bearing temperature remaining within limits while the winding temperature rose weakens H4.
- The absence of any motor protection relay operation and the healthy MCC status weaken H2 and H5.
- Drum level, furnace pressure, ID fan and SA fan status all normal at the moment of the trip exclude H6.
- The auto-start command with no run feed-back is direct support for H7 as a second, independent failure that turned a single-fan trip into a boiler trip.
- The physically choked cooling air passages confirm H1.

6.1 Most discriminating evidence

The most discriminating evidence is the divergence between winding temperature and motor current. If the motor were overloaded, current would rise with temperature; if the fault were electrical, protection would operate first. Winding temperature rising alone, at normal current and normal bearing temperature, identifies loss of the cooling air path specifically.

7. How competing hypotheses were ruled out

- H2 (electrical fault) ruled out - no motor protection relay operated, insulation resistance tested satisfactory after cooling, and the trip came from the thermal protection.

- H3 (high outlet pressure overload) ruled out - PA outlet pressure was well below the 1800 mmwc trip setting and all dampers were in their normal positions.
- H4 (bearing failure) ruled out - motor bearing temperature stayed within limits and bearing condition was found satisfactory on inspection.
- H5 (MCC fault) ruled out - MCC bus voltage and the incomer were healthy, and PA fan-B received its start command from the same MCC.
- H6 (consequential interlock trip) ruled out - all interlocked parameters were within their normal bands at the moment of the trip.
- H7 (standby auto-start failure) confirmed as a real, independent second failure and treated as a separate finding, not as an alternative to H1.

8. Final failure mechanism determined

Two independent failures in sequence. First, the PA fan-A motor overheated because its cooling air passages were fouled with coal dust, so the heat generated in the windings could not be removed, and the thermal protection tripped the fan. Second, the standby PA fan did not start because a defective run feed-back contact aborted the auto-start sequence - a hidden failure that was only revealed on demand. The combination removed all fluidising air, so the bed defluidised, the coal feeders tripped on the designed interlock, hot bed material migrated into the PA lines and the boiler tripped on low furnace draught.

9. Operational and maintenance context relevant to root cause

- The boiler house environment is dust-laden and motor cooling passages foul rapidly, but motor cleaning was not on a defined frequency.
- Motor winding temperature trends were available but were not reviewed until the alarm point.
- Standby PA fan auto-start was never proof-tested; standby availability was assumed rather than demonstrated - a classic hidden failure.
- The coal feeder trip on PA fan trip is a correct and necessary protection, but there was no accompanying automatic action to protect the bed (for example, controlled slumping, damper closure or burner support).
- No procedure existed for the immediate protection of PA lines against hot material migration on total loss of primary air.
- Bed material re-charging after a defluidisation event is time-consuming, and no pre-staged bed material stock was maintained.

10. Final root cause

Physical root cause: PA fan motor winding overheating from coal-dust fouling of the motor cooling air passages. Compounding root cause: unavailability of the standby PA fan due to a defective run feed-back contact. Systemic root cause: neither failure was detectable by the existing maintenance regime - there was no scheduled motor cleaning driven by the known dusty environment, and no periodic proof test of standby auto-start, so both a wear-out failure and a hidden failure were allowed to coincide.

11. Corrective actions to be taken

- Immediate: rectify the electrical fault if any, check the cause of the boiler trip and normalise, and check fan and motor bearing temperatures before restarting.
- Immediate: restart the PA fan only after checking the cause of tripping and taking corrective action.
- Immediate: clear the PA lines of migrated hot bed material before restarting fluidisation, and re-charge bed material as required.
- Immediate: ensure all DCFs are stopped if they have not tripped, and check PA lines for jamming and clear them accordingly.
- Short term: clean the motor cooling air passages, test insulation resistance, and repair or replace the PA fan-B run feed-back contact; prove the auto-start by test.

- Short term: on any PA fan trip, see that the standby fan is started - manually if the auto-start fails - and increase the PA header pressure to restore fluidisation quickly.
- Medium term: schedule periodic motor cleaning and thermographic checks based on the dust loading of the boiler house, and trend winding temperature as condition data.
- Medium term: introduce periodic proof testing of all standby auxiliary auto-starts (PA, SA, ID, BFP), since these are hidden failures that only appear on demand.
- Long term: review whether an automatic bed-protection sequence (damper closure and controlled slumping) should follow a total loss of primary air, and maintain a pre-staged bed material stock to shorten restoration.

Cross-cutting findings across the eighteen cases

Read together, the cases show a small number of systemic weaknesses recurring behind very different physical failures. These are the areas where corrective effort yields the widest benefit:

- Hidden failures in standby equipment. Standby BFP auto-start, standby PA fan auto-start, the PRDS dumping valve and the light-up oil system all failed only when demanded. None was proof-tested. Periodic functional testing of standby and protective functions is the single highest-value addition to the maintenance programme.
- Operating outside the design envelope after a change. Fuel calorific value, ash content, sulphur content, load level and heater availability all changed without a matching change to air flow, feeder speed, bed inventory or load limit. A documented re-setting procedure for each such change would have prevented Cases 3, 6, 7, 8 and 13.
- Degradation mechanisms that were predictable but unmeasured. Tube erosion, creep, oxide scale and cold-end corrosion are wear-out mechanisms with known drivers. None was being measured or trended, so each ran to failure.
- Single points of failure. One MCC bus for all six feeders, one deaerator for both feed pumps, and a single-fan capacity margin insufficient for the operating load each turned a component failure into a unit trip.
- Work control and restoration discipline. A closed cooling water valve, a throttled damper and a defeated purge permissive each caused or enabled a serious event. Permit-to-work, tagging and post-maintenance restoration checks address all three.
- Alarms without defined actions. High bed temperature, high SH metal temperature, low deaerator level and low drum level were all annunciated well before damage. Where an alarm carries no mandated action and no interlock, it does not protect the plant.